

Supplementary Information for

Leveraging Large Language Models for Redundancy-Aware Pathway Analysis and Deep Biological Interpretation

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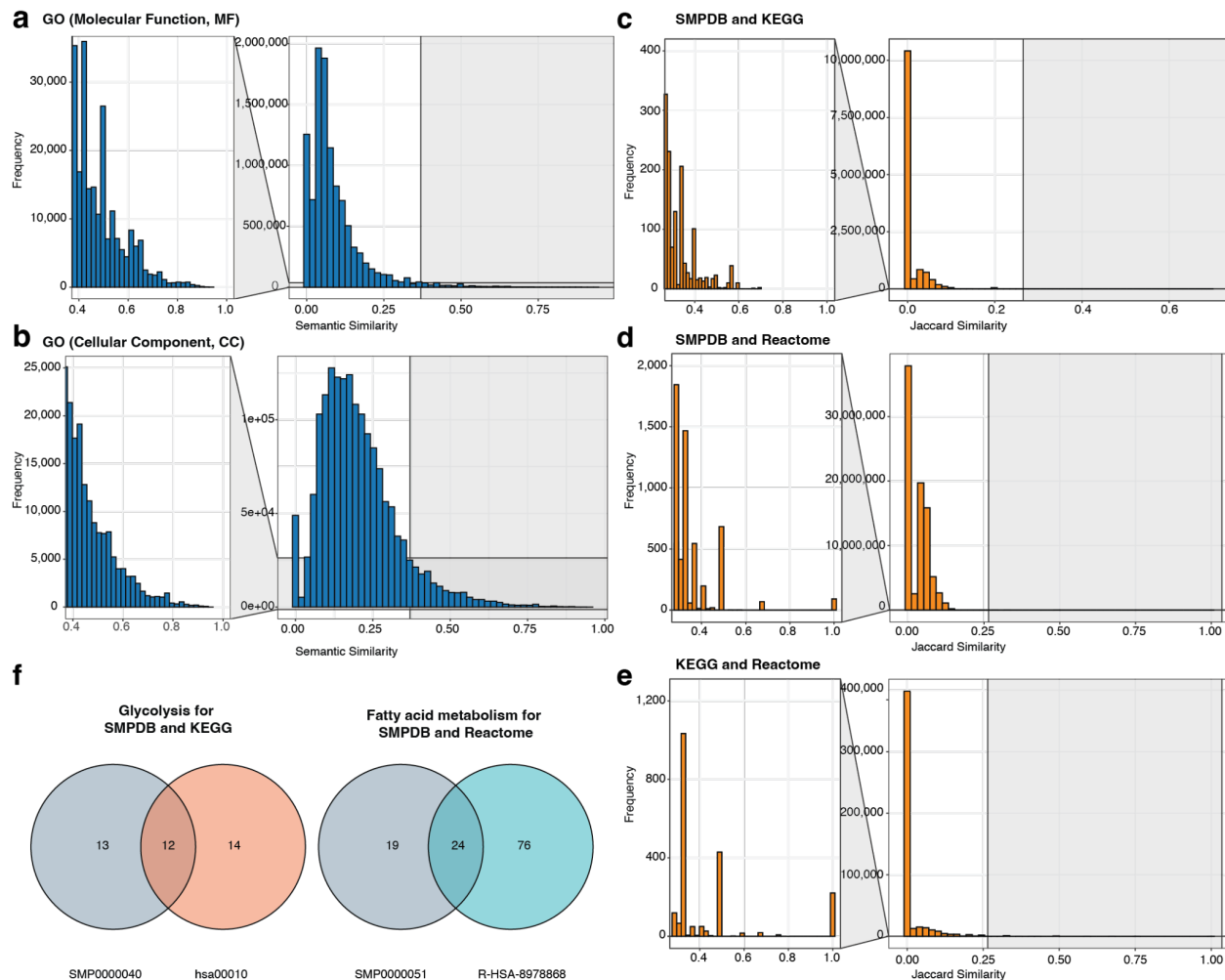
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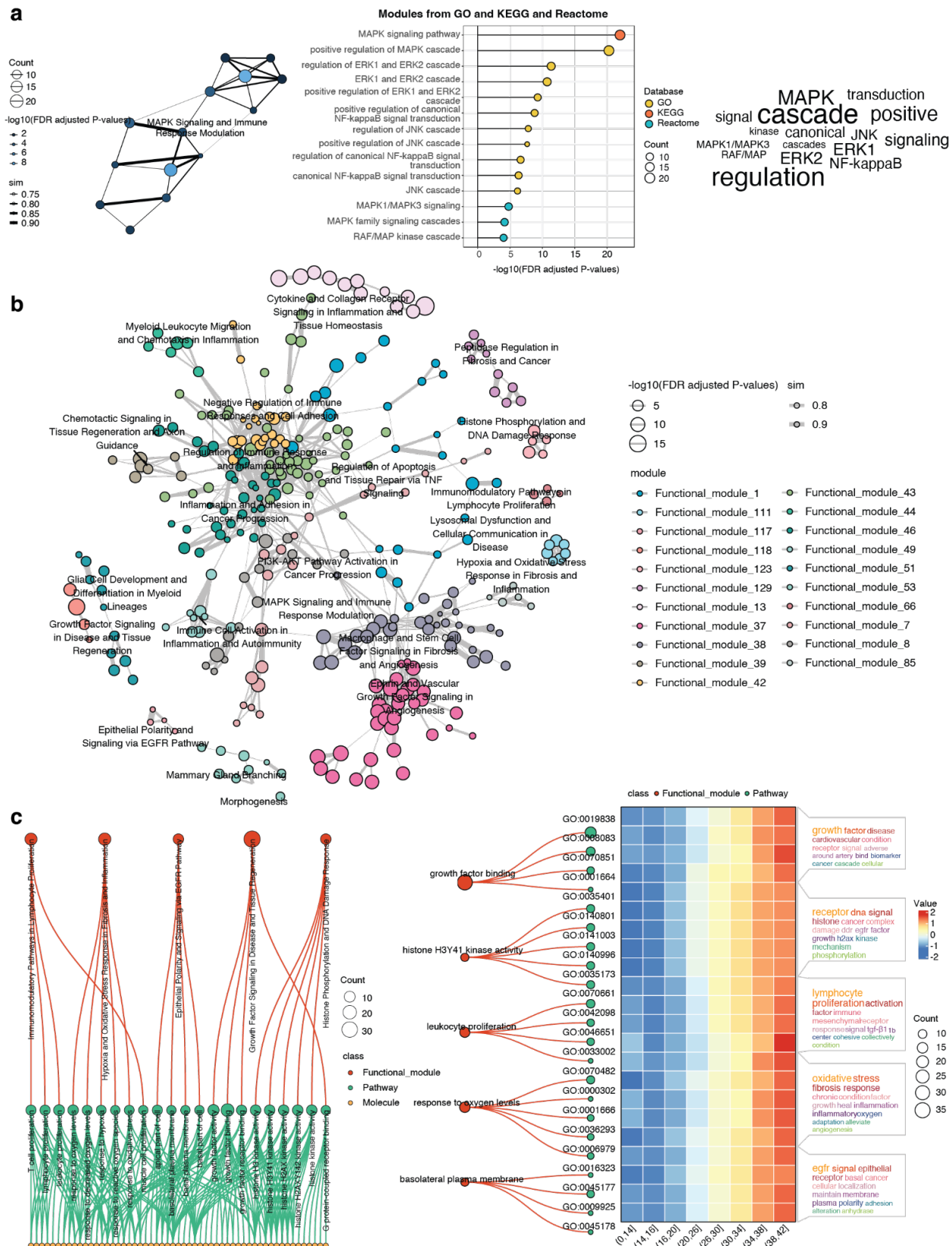
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Supplementary Figure 1. Redundancy within Gene Ontology and across metabolic pathway databases. **a,b**, Histograms showing the distribution of pairwise Wang similarity scores for Gene Ontology (GO) terms within the molecular function (a) and cellular component (b) subontologies. A substantial proportion of GO term pairs exhibit high semantic similarity, indicating significant redundancy within the GO database. **c–e**, Histograms of pairwise Jaccard similarity scores for human pathways across different metabolic databases: SMPDB vs. KEGG (c), SMPDB vs. Reactome (d), and KEGG vs. Reactome (e). The presence of many pathway pairs with high similarity scores indicates extensive cross-database redundancy. **f**, Two representative biological processes represented by pathways from different databases that share many metabolites. Venn diagrams illustrating the overlap in molecular components for representative biological processes. Left: Glycolysis pathways from SMPDB (SMP000040) and KEGG (hsa00010). Right: Fatty acid metabolism pathways from SMPDB (SMP000051) and Reactome (R-HSA-8978868).



Supplementary Figure 2. Representative MAPA outputs generated from the package's built-in example dataset. **a**, Module detail view combining the intra-module network, pathway enrichment summary (GO/KEGG/Reactome), and a keyword cloud. **b**, Similarity network of functional modules (≥ 3 nodes) illustrating relationships among modules. **c**, Left, a multipartite graph linking molecules, pathways, and functional modules; right, an expression heatmap integrated with the relationship network and keyword clouds, jointly representing functional themes and expression patterns.



Module Annotation for Pathway Enrichment Result Analysis

mapa by The Shen lab

2025-08-10

Introduction

mapa (version 0.1.30): Created in 2021 by The Shen lab

Parameters

package_name	function_name	parameter	time
mapa	enrich_pathway()	query_type: gene	2025-08-10 15:37:23
mapa	enrich_pathway()	database: go,kegg,reactome	2025-08-10 15:37:23
mapa	enrich_pathway()	save_to_local: FALSE	2025-08-10 15:37:23
mapa	enrich_pathway()	pathresult	2025-08-10 15:37:23
mapa	enrich_pathway()	go.org.db.org.Hs.ag.db	2025-08-10 15:37:23
...	...	...	...

Object information

```
#> -----
#> Analysis method: enrich_pathway
#> -----
#> Variable information:
#> 119 features/markers in total
#> -----
#> Enrichment results and modules of genes:
#> -----
#> 1025 GO terms with p.adjust < 0.05
#> No GO modules
#> -----
#> 49 KEGG pathways with p.adjust < 0.05
#> No KEGG modules
#> -----
#> 48 Reactome pathways with p.adjust < 0.05
#> No Reactome modules
#> -----
#> Enrichment results and modules of metabolites:
#> -----
#> Functional modules:
#> 132 Functional modules
#> -----
#> LLM module interpretation:
#> 27 LLM interpreted modules
#> -----
#> Processing information
#> 4 processings in total
#> enrich_pathway -----
#> Package Function.used Time
#> 1 mapa enrich_pathway() 2025-08-10 15:37:23
#> merge_pathways -----
#> Package Function.used Time
#> 1 mapa get_bloembedsim() 2025-08-10 15:45:13
#> merge_modules -----
#> Package Function.used Time
#> 1 mapa get_bloembedsim() and merge_pathways_bloembedsim() 2025-08-10 15:59:51
#> l1m_interpret_module -----
#> Package Function.used Time
#> 1 mapa l1m_interpret_module() 2025-08-10 16:25:23
```

Bar plot shows the top 10 modules/pathways

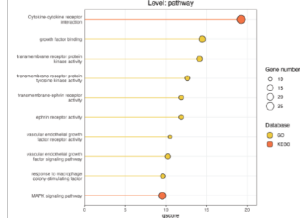


Figure 1: Barplot of the enriched pathways/modules (p.adjust.cutoff = 0.05, count.cutoff = 5).

Module network

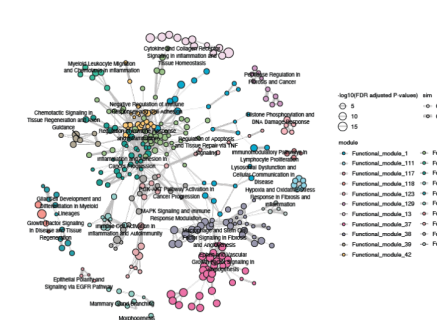
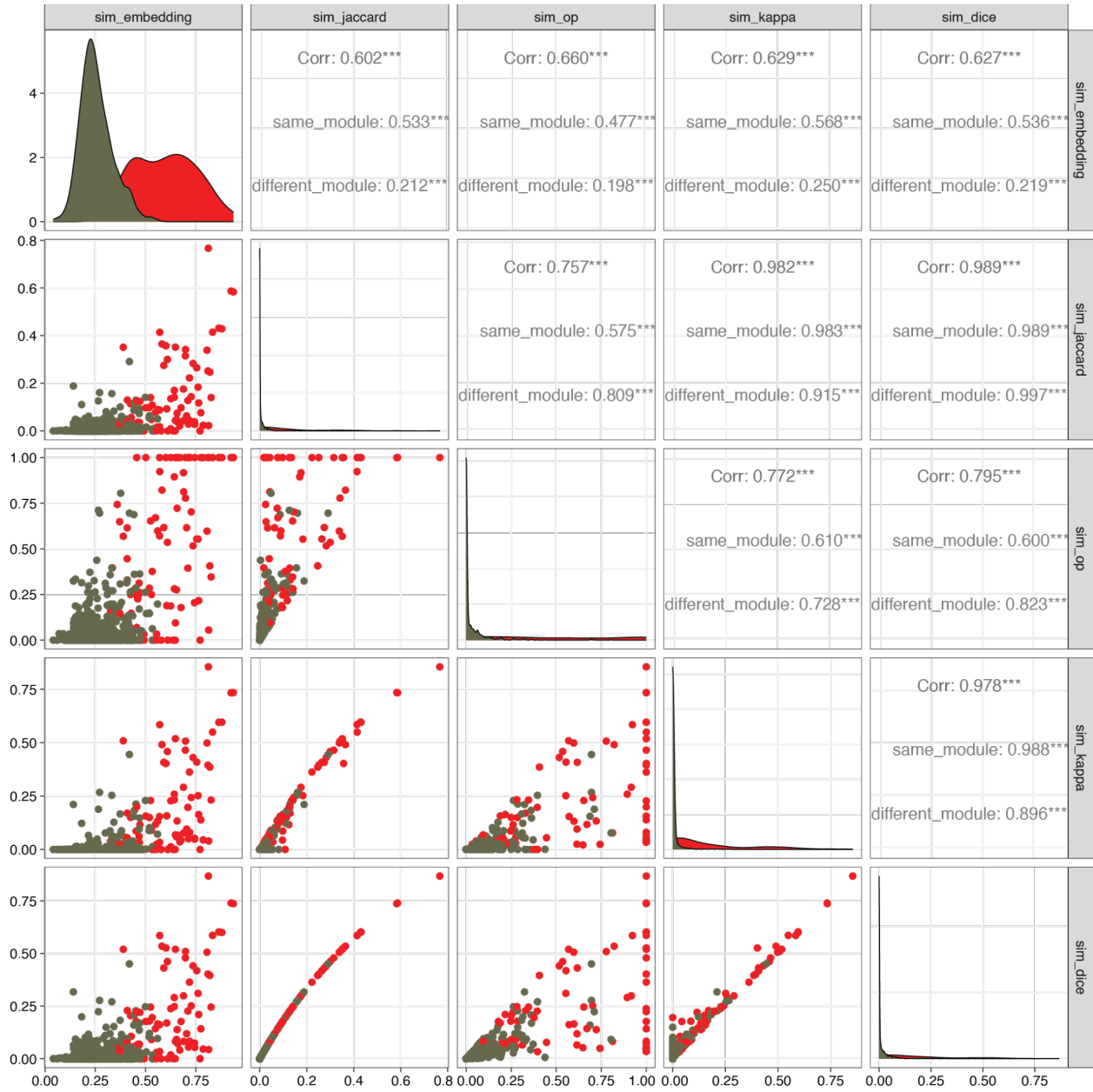


Figure 2: Module networks of the enriched pathways/modules.

Interpretation of functional modules

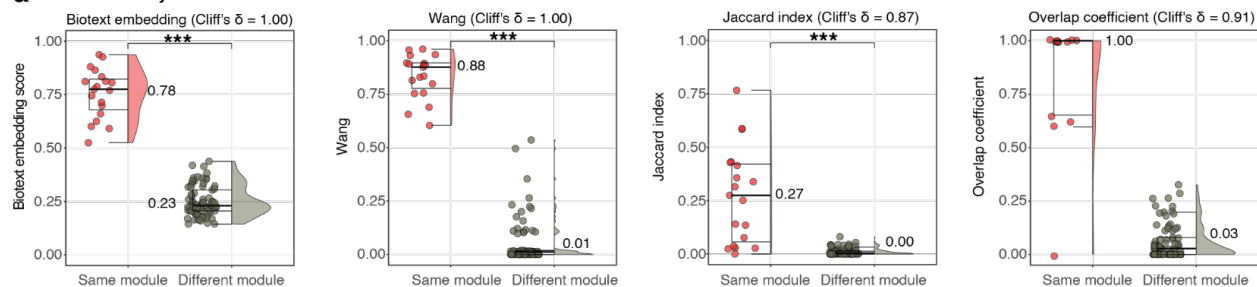
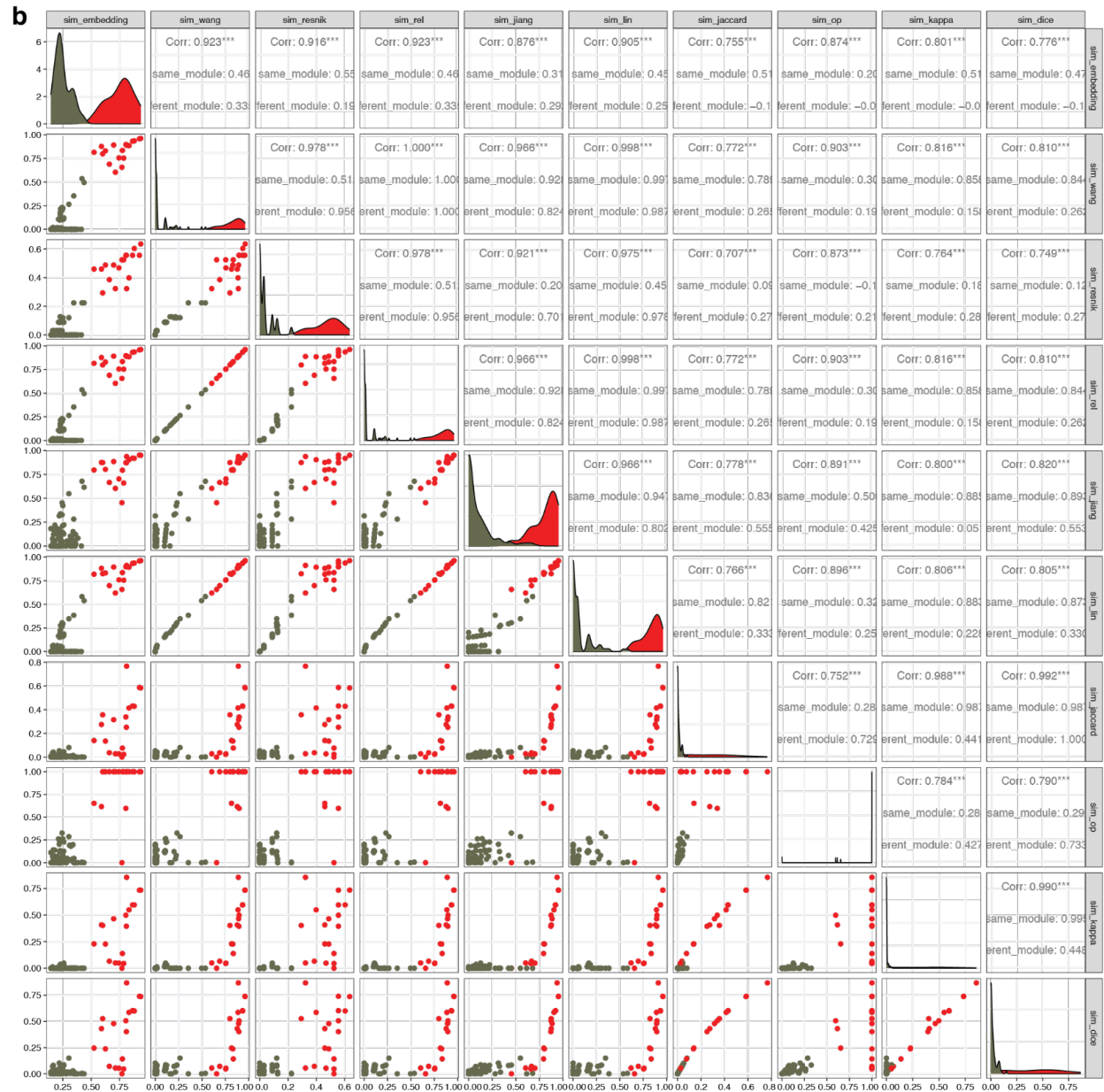
module	llm_interpreted_name	llm_summary	llm_phenotype_analysis	confidence_score
Functional_module_13	Cytokine and Collagen Receptor Signaling in Inflammation and Tissue Homeostasis	This module investigates the intricate signaling mechanisms involving cytokines and collagen receptors, highlighting their roles in inflammation and tissue homeostasis. Central to this module are pathways related to cytokine-cytokine receptor interactions and ...	NA	0.92
Functional_module_118	Growth Factor Signaling in Disease and Tissue Regeneration	This module centers around the roles of growth factors and their receptors in regulating cellular growth, differentiation, and survival, with implications in various diseases including cancer and cardiovascular ...	NA	0.90
...	...	...	...	...

Supplementary Figure 3. Example MAPA structured report. The report compiles: (1) package version, and timestamps; (2) input parameters; (3) object information summarizing methods, databases queried, and processing steps; (4) bar plot of the top 10 enriched pathways or representative pathways from functional modules; (5) module network summarizing relationships among functional modules and enriched pathways; (6) an interpretation table listing module ID, functional name, summary, module-phenotype analysis, and confidence score generated by large language model (LLM).

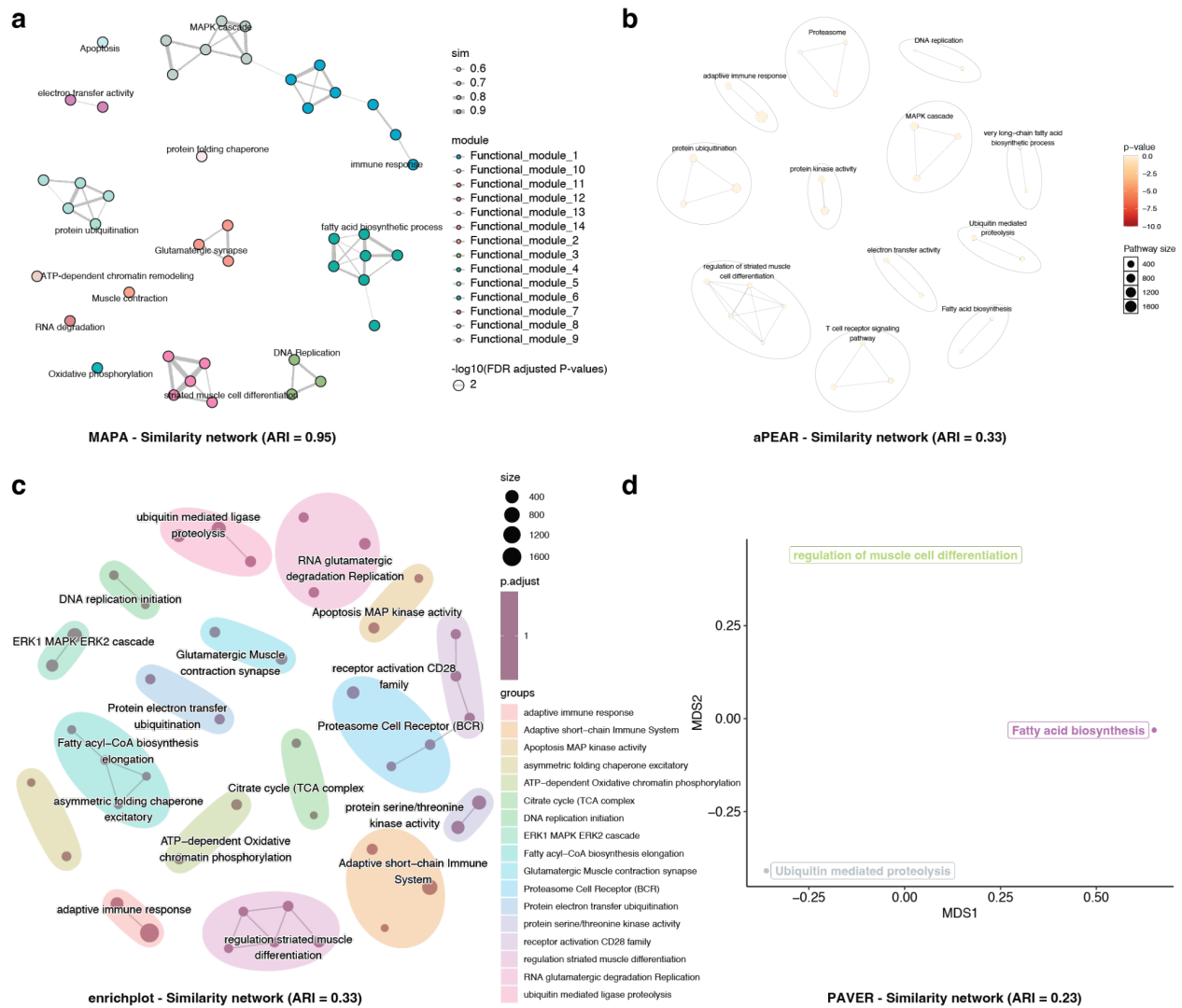


Supplementary Figure 4. Pairwise correlation among different similarity metrics. The scatterplot matrix compares five component-based similarity scores. Diagonal plots show the score distribution for each method, distinguishing between pairs from the same module (red) and different modules (dark grey). Off-diagonal plots show the pairwise correlations between metrics. Pearson correlation coefficients (Corr) are shown in the upper triangle, with asterisks denoting significance (***) p-values < 0.001). The Jaccard index, Cohen's Kappa, and Dice coefficient are highly correlated ($r > 0.97$).

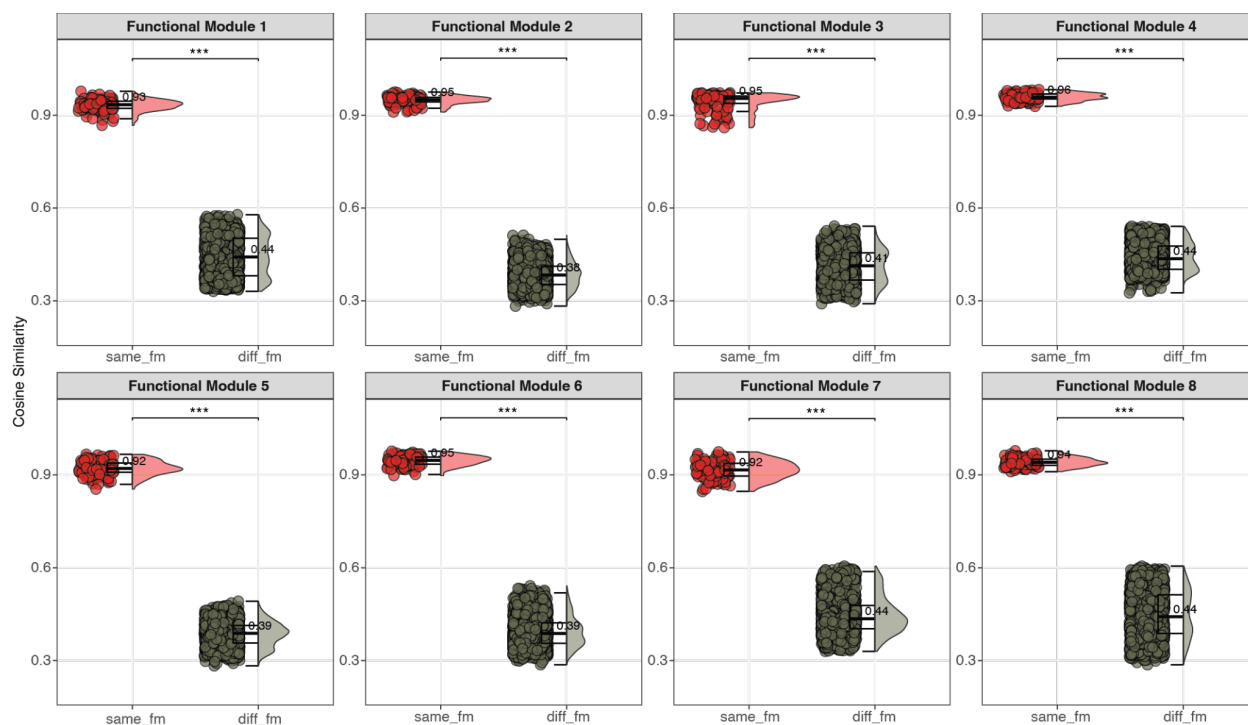
a GO terms only

**b**

Supplementary Figure 5. Discriminative power of similarity metrics specifically for GO terms. **a**, Comparison of similarity scores for within-module (red) versus between-module (dark gray) GO-term pairs. Cliff's delta (δ) quantifies separation. Biotext embedding achieves perfect discrimination ($\delta = 1.00$, $p < 0.001$), outperforming Jaccard ($\delta = 0.87$) and overlap coefficient ($\delta = 0.91$). Wang semantic similarity also attains $\delta = 1.00$ for computable GO term pairs but cannot be evaluated across GO sub-ontologies, resulting in substantial missing values and highlighting a practical limitation for cross-ontology comparisons. **b**, Pairwise correlation among similarity metrics. The scatterplot matrix compares biotext embedding, semantic similarity metrics specifically for GO terms (Wang, Resnik, Rel, Jiang, and Lin), and component-based similarity scores (Jaccard, overlap coefficient, Cohen's Kappa, and Dice coefficient). Diagonal plots show the score distribution for each method, distinguishing within-module pairs (red) and between-module pairs (dark gray). Off-diagonal panels show pairwise relationships; the upper triangle reports Pearson correlations (Corr) with asterisks for significance (***) p -values < 0.001 . The similarity scores generated by Wang, Resnik, Rel, Jiang, and Lin methods are highly correlated ($r > 0.97$). The Jaccard index, Cohen's Kappa, and Dice coefficient are highly correlated ($r > 0.98$).

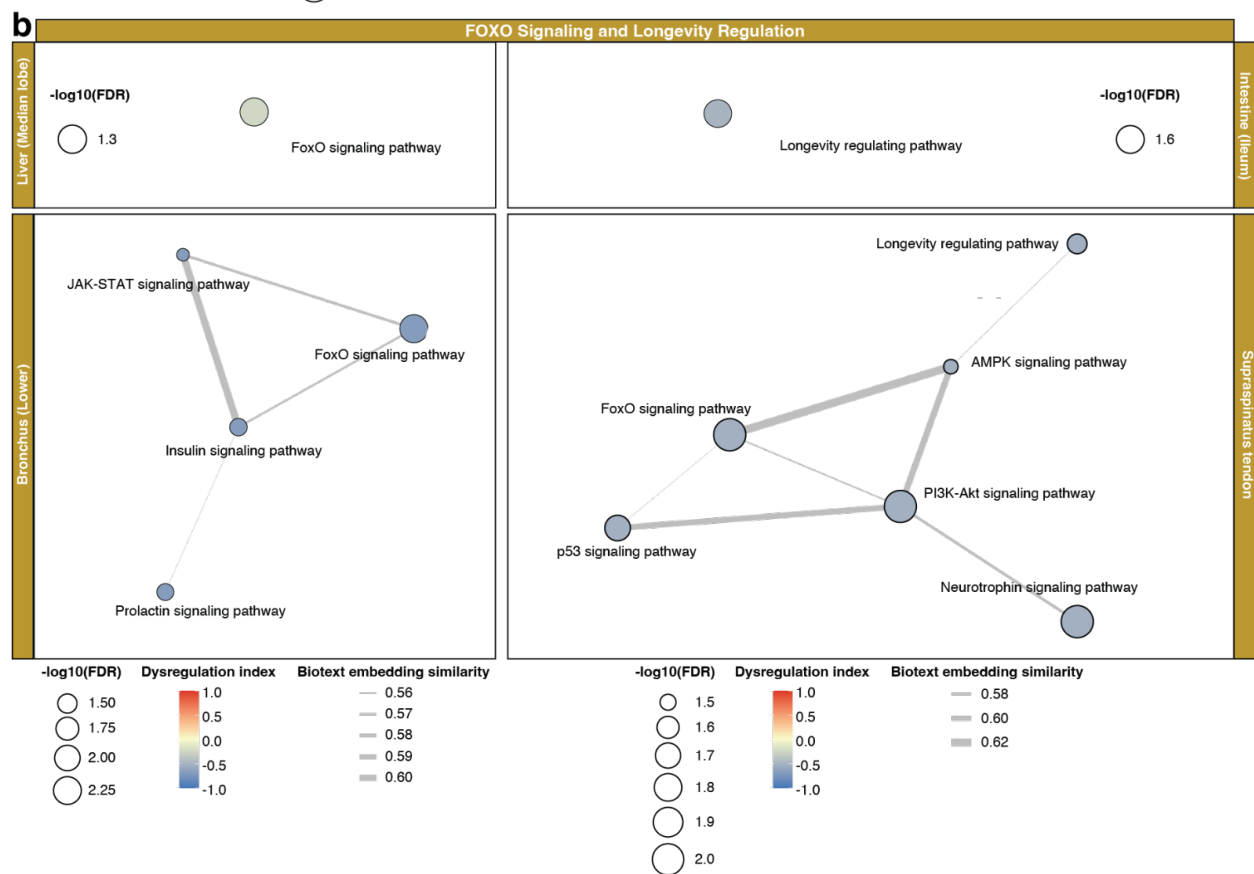
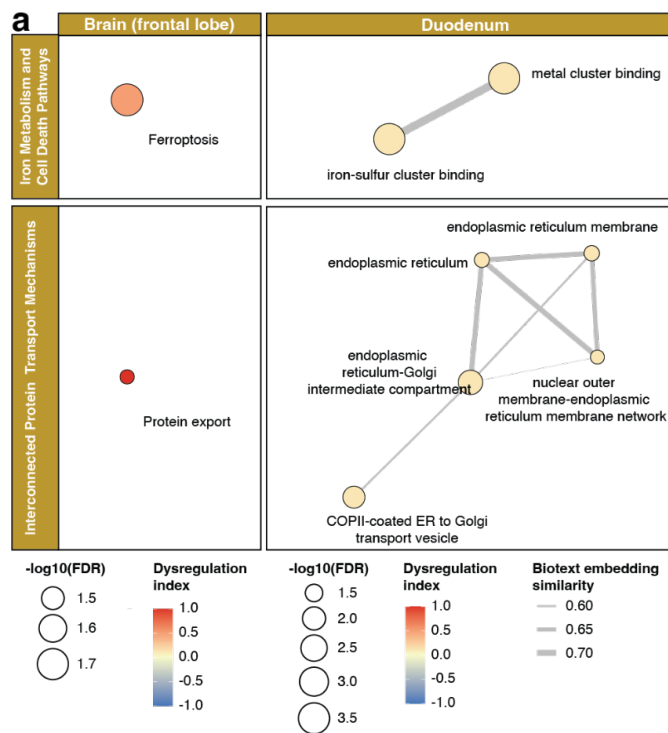


Supplementary Figure 6. Benchmarking MAPA against other pathway redundancy analysis tools on the curated dataset 1. Comparison of similarity networks from four tools for a set of pathways. Performance was measured by the adjusted rand index against a ground truth. **a**, MAPA similarity network achieving the highest accuracy (ARI = 0.95). Nodes (pathways) are grouped into distinct functional modules. Edges represent biotext similarity scores (> 0.55), and module labels are names generated by LLM. **b**, aPEAR similarity network showing significantly lower performance (ARI = 0.33). Edges represent the Jaccard index, and modules are labeled using the name of the most connected pathway in each module. **c**, enrichplot similarity network also demonstrates low accuracy (ARI = 0.33). Edges represent a Jaccard index > 0.2 , and modules are labeled with the most specific shared terms. **d**, PAVER multi-dimensional scaling plot showing the lowest accuracy (ARI = 0.23). Nodes represent entire functional modules, which are labeled with the name of the most representative pathway with the highest cosine similarity to the cluster's average embedding.



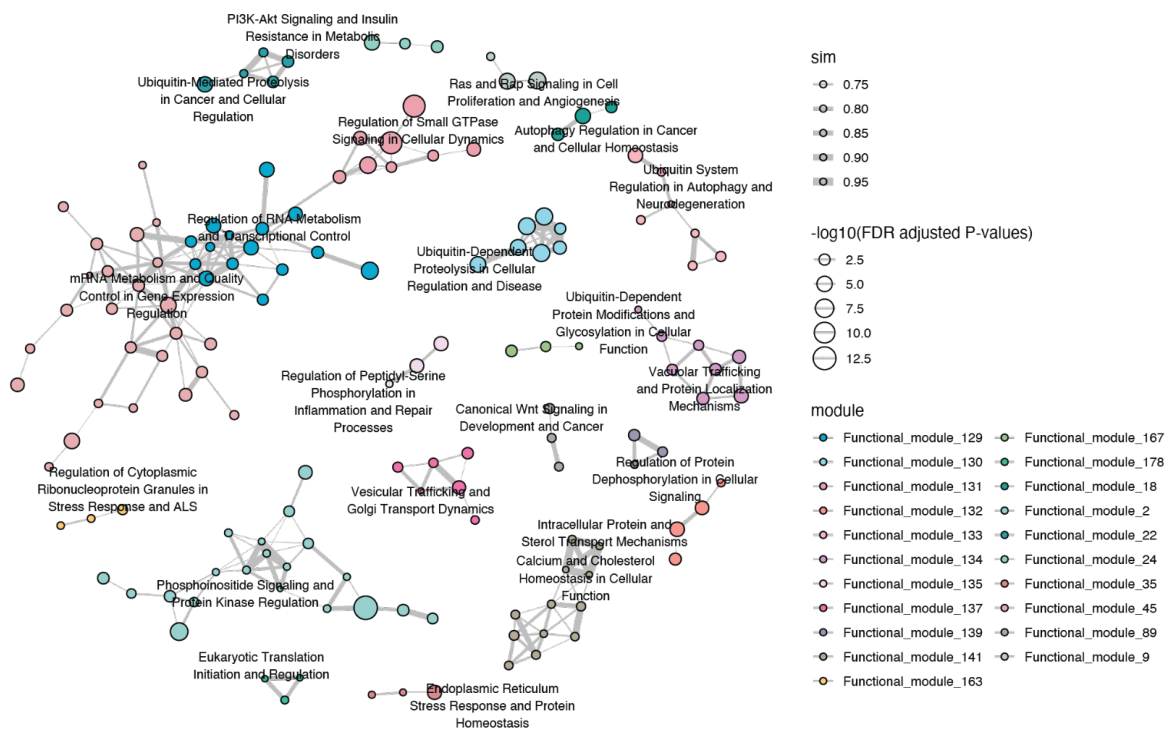
Supplementary Figure 7. Reproducibility for MAPA’s performance across independent runs on curated data.

The curated dataset 1 was processed through MAPA’s clustering and LLM interpretation pipeline for 15 independent runs. Consistent clustering was observed across runs (ARI = 1, resulting in 14 functional modules). LLM-generated functional summaries were obtained for the 8 functional modules containing more than one pathway. For each of these modules, the pairwise cosine similarity of functional annotations across the 15 runs (same_fm) was compared to the cosine similarity of annotations between different functional modules across all runs (diff_fm). High cosine similarity (> 0.91) was observed for annotations of the same functional module across different runs, indicating strong reproducibility. In contrast, annotations between different functional modules exhibited low similarity. Statistical significance was assessed using a Wilcoxon test (***) p-value < 0.001).

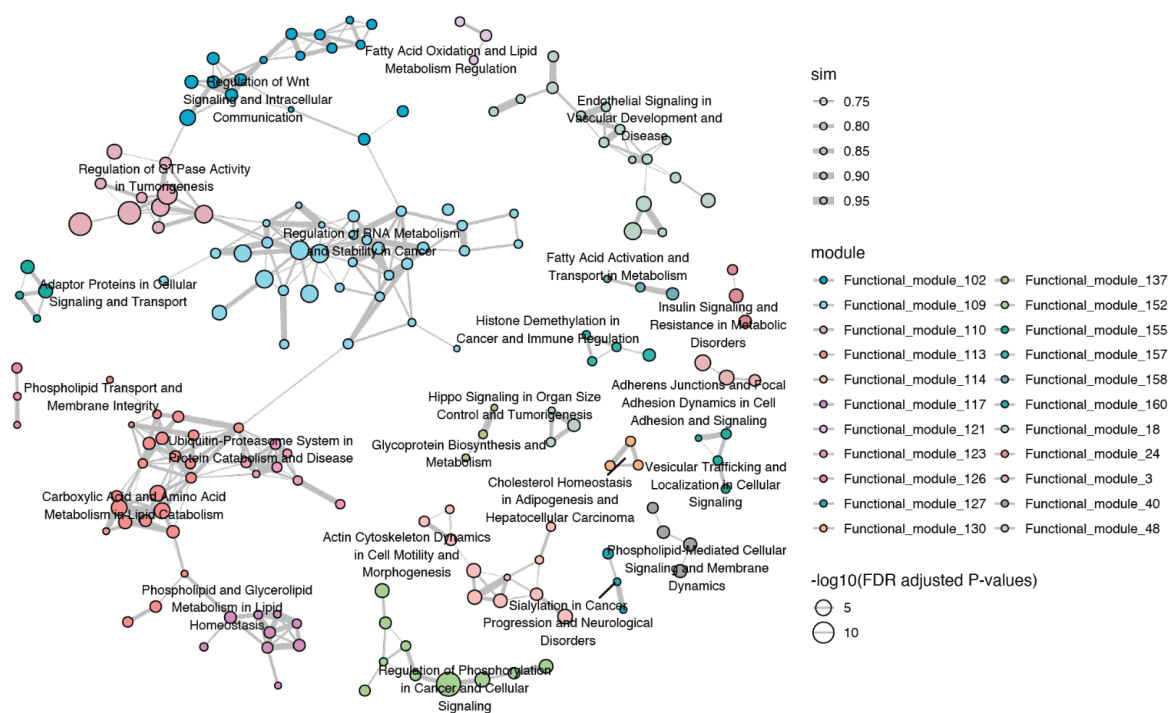


Supplementary Figure 8. Shared age-related functional modules across diverse tissues. Functional modules that are similarly regulated during aging in different tissues. Node size corresponds to significance ($-\log_{10}(\text{FDR})$), and color indicates upregulation (red) or downregulation (blue). **a**, Upregulated functional modules shared between the brain (frontal lobe) and intestine (duodenum). These modules are primarily involved in iron metabolism, cell death (ferroptosis), and protein transport mechanisms. **b**, Downregulated functional modules related to FOXO signaling and longevity regulation. These changes are consistently observed across multiple tissues, including the liver (median lobe), intestine (ileum), bronchus (lower), and supraspinatus tendon.

a

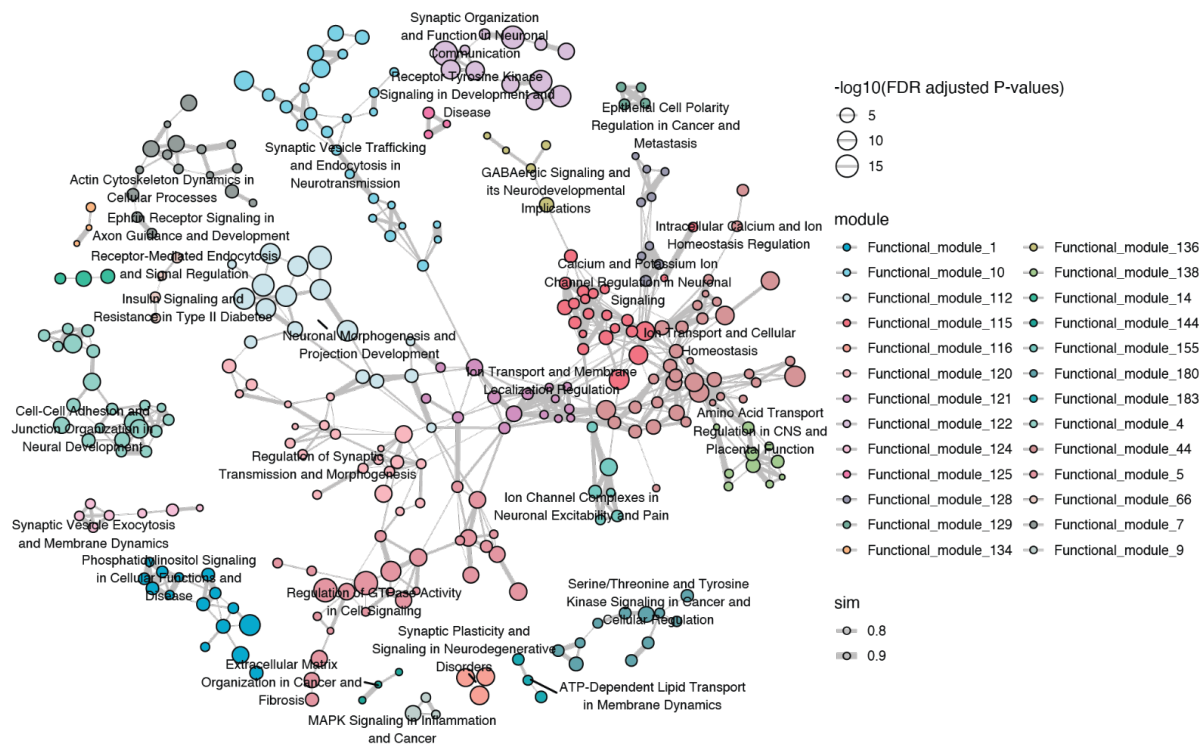


b

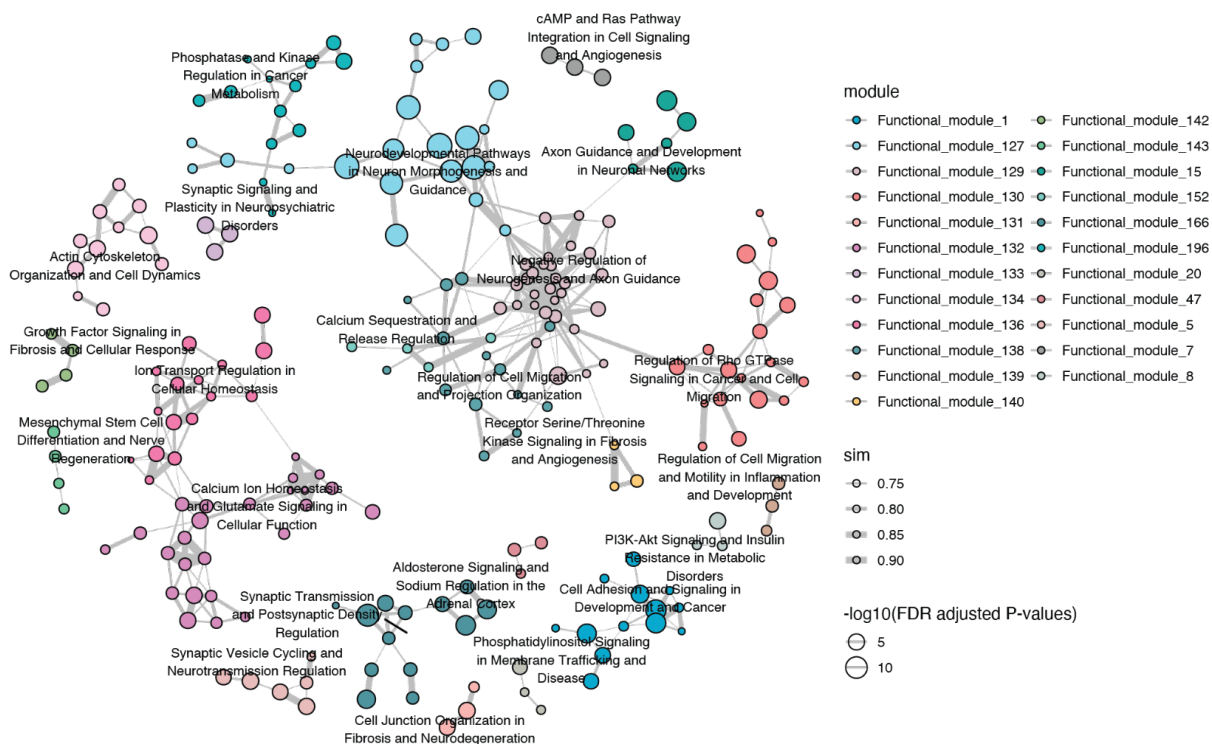


Supplementary Figure 9. MAPA reveals biologically meaningful functional networks in age-dependent liver gene expression profiles. Functional module networks of upregulated (a) and downregulated (b) differentially expressed genes from an age-dependent liver snRNA-seq dataset¹. The networks were generated using the MAPA workflow to provide a biologically meaningful view of the functional relationships between pathways. Each node represents an enriched pathway, where node size is proportional to the statistical significance ($-\log_{10}$ FDR-adjusted p-value) and the edge width reflects the biotext similarity between connected pathways. **a**, Functional module network for upregulated genes (cluster U). MAPA successfully corroborates findings from the original study, identifying key functional modules associated with inflammatory and immune processes, such as “Regulation of Peptidyl-Serine Phosphorylation in Inflammation and Repair Processes” and “Ubiquitin-Dependent Proteolysis in Cellular Regulation and Disease.” **b**, Functional module network for downregulated genes (cluster D). Beyond replicating the original findings, MAPA identified a key module, “Carboxylic Acid and Amino Acid Metabolism in Lipid Catabolism,” which functionally connects lipid and amino acid metabolic processes. This highlights a more integrated view of metabolic dysregulation in aging, showcasing MAPA’s power to reveal deeper functional connections missed by conventional analyses.

a

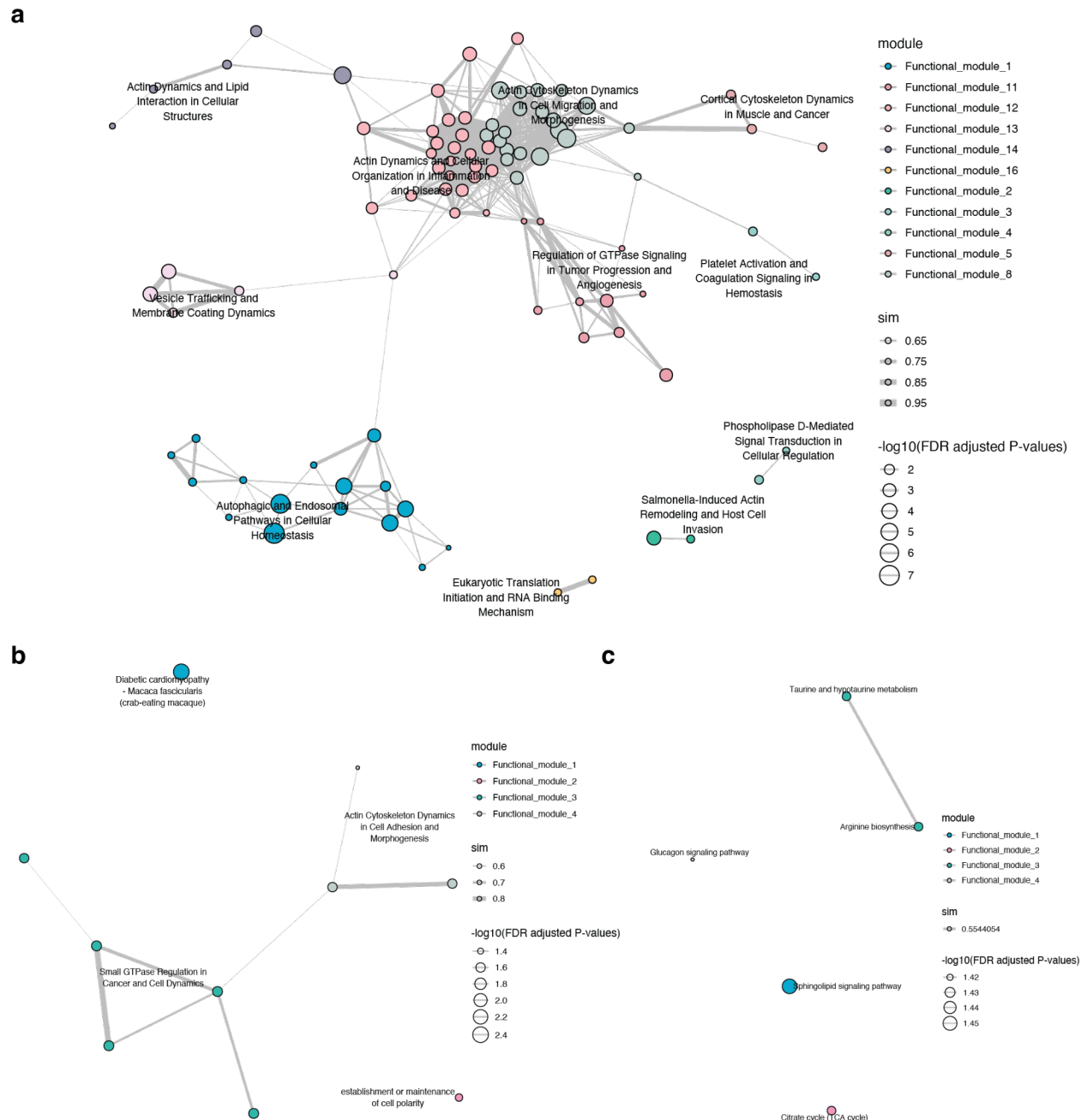


b

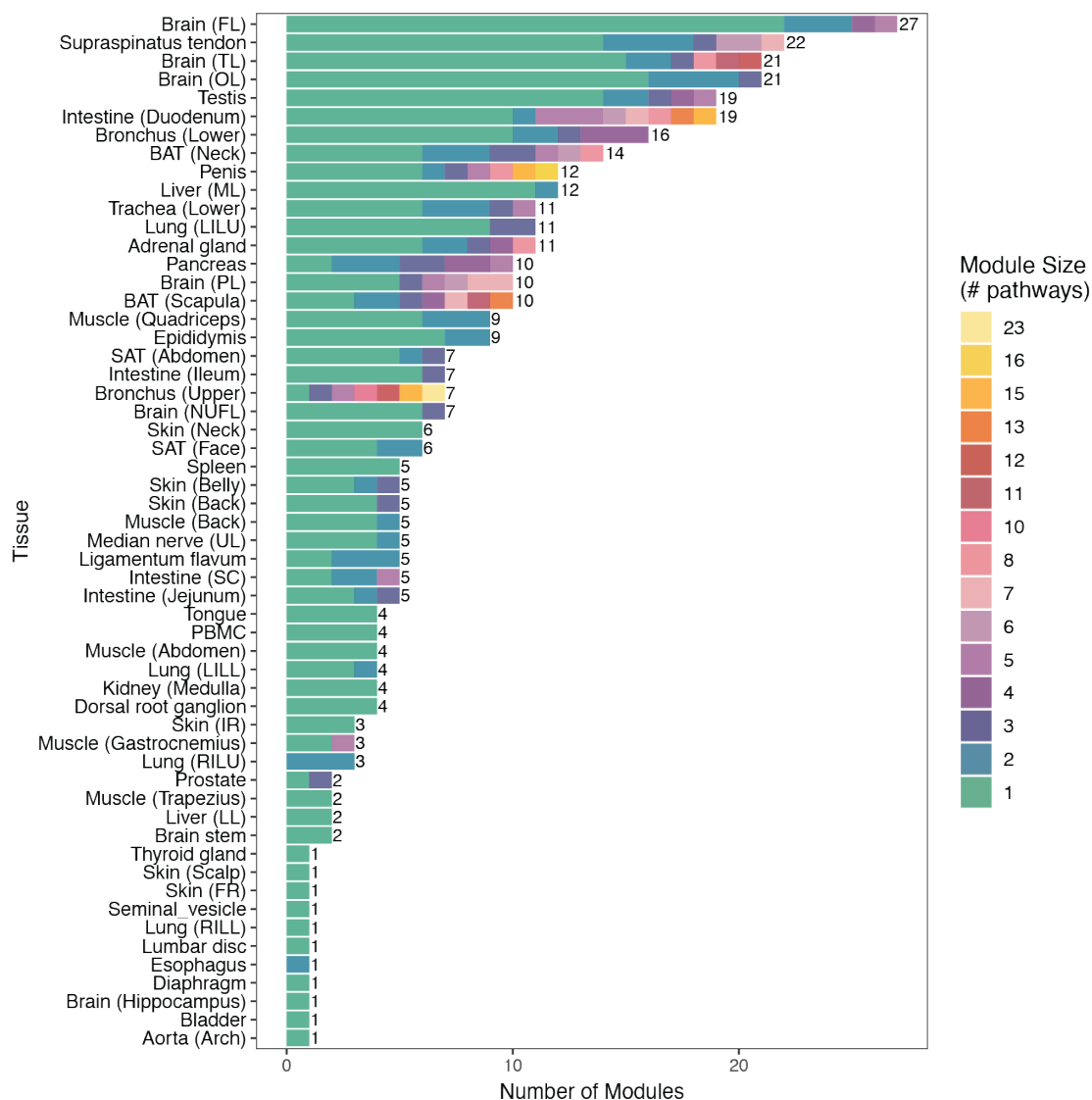


Supplementary Figure 10. MAPA identifies biologically meaningful functional modules in the aging brain.

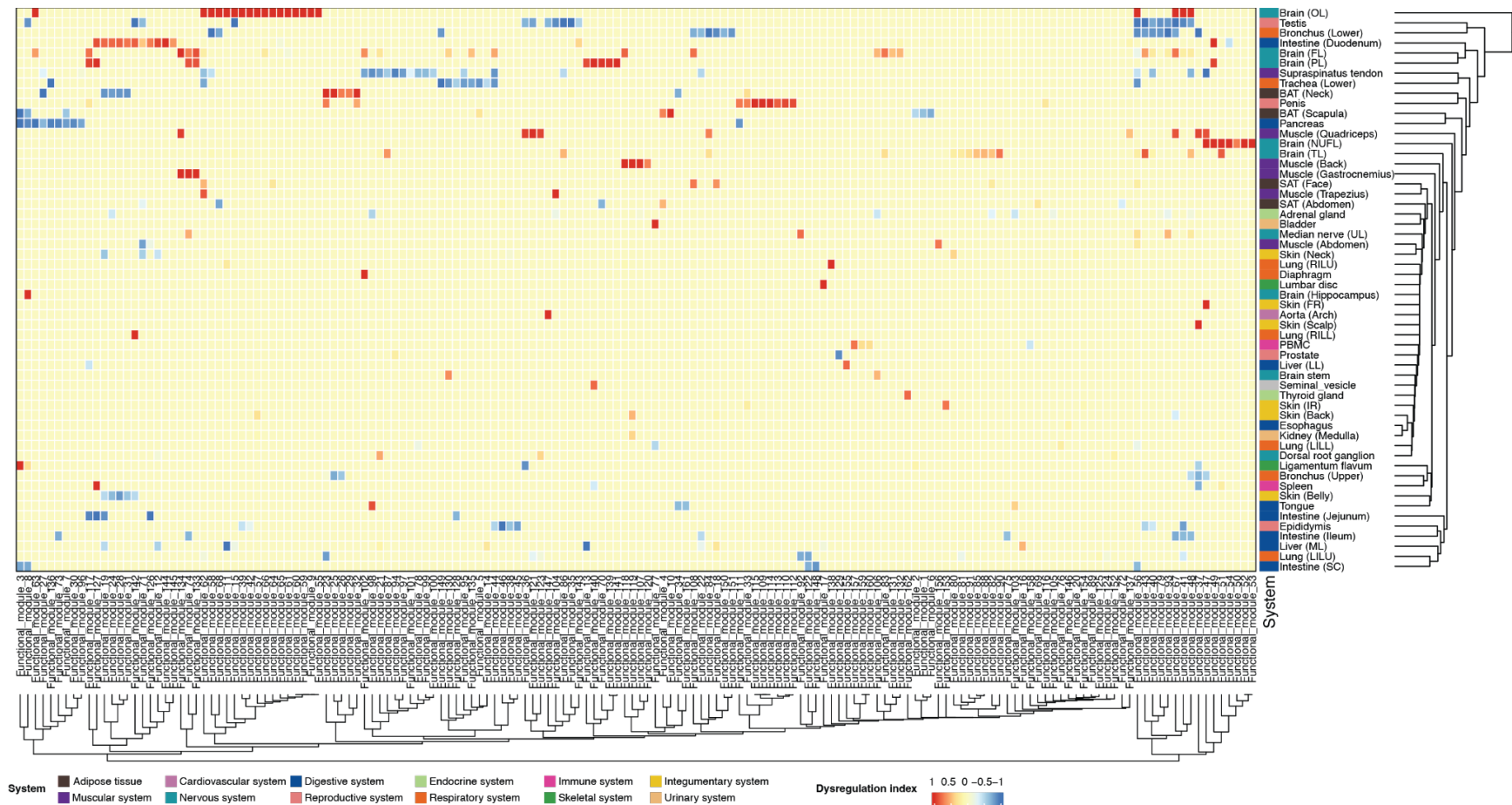
Functional module similarity networks of upregulated (a) and downregulated (b) genes from an age-dependent brain snRNA-seq dataset¹, generated using the MAPA workflow. **a**, Functional module network for upregulated genes (cluster U). The functional modules identified by MAPA support the findings of the original study. For instance, it uncovers a neuroinflammatory module (“MAPK Signaling in Inflammation and Cancer”) that reflects known age-related changes in the brain². **b**, Functional module network for downregulated genes (cluster D). While consistent with previous findings on synaptic decline, MAPA synthesizes related concepts into a more informative module: “Synaptic Transmission and Postsynaptic Density Regulation.” This offers a more integrated view than a singular term like “neurotransmitter secretion,” as it explicitly links the process of synaptic transmission with the structural organization of the postsynaptic density, providing a more nuanced biological insight.



Supplementary Figure 11. MAPA identifies functional modules from age-related plasma proteomics and metabolomics data. MAPA was applied to lists of age-related plasma proteins and metabolites¹ to construct functional module similarity networks. Network visualization is consistent with previous figures. **a**, Functional module network for downregulated proteins (cluster D). MAPA identified a key module, “Autophagic and Endosomal Pathways in Cellular Homeostasis,” which integrates pathways related to cellular protein and organelle degradation. This provides a unified view of declining cellular clearance mechanisms with age. **b**, The network for upregulated proteins (Cluster U). **c**, The network derived from both upregulated and downregulated metabolites.



Supplementary Figure 12. Distribution of functional module count and size across tissues. The stacked bar chart shows the number and size of functional modules identified in 56 tissues. Each bar represents a tissue, with its length indicating the module count. The color corresponds to the module's size (number of pathways) as defined in the legend.



Supplementary Figure 13. Heatmap of Dysregulation Index for functional modules generated from age-dependent differentially expressed genes identified from bulk RNA-seq data across tissues. The heatmap displays the Dysregulation Index for 396 functional modules across 56 tissues (rows), with both axes organized by hierarchical clustering. The index color scale indicates each module's regulatory trend during aging, from downregulated (blue, -1) to upregulated (red, +1). Tissues are annotated by the physiological systems on the right, and functional modules are grouped into 162 meta-modules at the bottom. The visualization highlights conserved patterns of aging across diverse biological tissues.

Supplementary Table 1. Comparison of MAPA with existing pathway clustering tools. This table compares the features of MAPA with other popular tools for clustering and annotating enriched pathways.

Tools	Input - can accept a simple list	Pathway similarity calculation			Clustering methods	Annotation methods	
		Component-based similarity	Semantic similarity	Embedding-based similarity		Module name	Module summary
enrichplot	No Requires pre-generated enrichment results from clusterProfiler.	Yes	Yes	No	A selection of standard algorithms (e.g., k-means, PAM) that require the user to pre-specify the number of clusters.	Yes Selects top words based on word frequency within pathway names.	No
aPEAR	No Requires pre-generated enrichment results from specific tools or in a strict format.	Yes	No	No	Limited choice of algorithms with fixed parameters (e.g., Markov clustering, hierarchical clustering with a fixed cutoff).	Yes Selects a single representative pathway based on network connectivity (PageRank, HITS) or statistical significance (max NES or min p-value).	No
PAVER	No Requires three separate inputs: enrichment results, a pathway ID map, and pre-computed pathway embeddings.	No	No	Yes	A single hierarchical clustering method	Yes Selects a single representative pathway based on the highest cosine similarity to the cluster's average embedding.	No
MAPA	Yes Accepts a direct list of	Yes	Yes	Yes	Offers a suite of hierarchical, graph-based, and binary cut	Yes Provides two	Yes LLM-powered

	gene, protein, or metabolite identifiers, automating the enrichment step.				clustering algorithms	approaches: (1) Novel LLM-powered interpretation for module naming, and (2) Traditional selection of the most significant pathway.	interpretation based on both database knowledge and current studies from PubMed literature
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Supplementary Table 2. The annotation results from five human experts.

Module	Expert #1	Expert #2	Expert #3	Expert #4	Expert #5
4	Module name: Glutamatergic synapse Module description: This module describes the neuron-to-neuron synapse that uses glutamate as a neurotransmitter, which results in excitatory postsynaptic potentials. Confidence score: 0.95	Module name: Excitatory neurotransmission via glutamate Module description: Excitatory neurotransmission via glutamate is a form of synaptic signaling in the mammalian central nervous system (CNS), in which glutamate serves as the principal excitatory neurotransmitter. At glutamatergic synapses, glutamate is packaged into synaptic vesicles in the presynaptic terminal and released into the synaptic cleft upon neuronal activation. It then binds to ionotropic glutamate receptors (iGluRs) on the postsynaptic membrane, leading to fast excitatory postsynaptic potentials. Confidence score: 0.95	Module name: Glutamatergic Excitatory Synapse Module Module description: This is a specialized glutamatergic synaptic transmission system featuring complete excitatory synapse machinery: glutamate release, ionotropic/metabotropic receptors, and glutamate-glutamine recycling. This module represents the fundamental excitatory signaling unit in the mammalian central nervous system. Confidence score: 0.95	Module name: Glutamatergic synapse pathway Module description: The overarching biological process is glutamatergic synaptic transmission, the principal excitatory signaling mechanism within the central nervous system, mediated by the neurotransmitter glutamate. The process begins with glutamate being packaged into synaptic vesicles within the presynaptic neuron. Upon neuronal activation, vesicles fuse with the presynaptic membrane, releasing glutamate into the synaptic cleft. Extracellular glutamate subsequently binds to postsynaptic receptors, including ionotropic (NMDA, AMPA, kainate receptors) and metabotropic glutamate receptors (mGluRs), propagating excitatory signals to downstream neurons. Precise regulation of glutamatergic signaling is essential for neuronal plasticity, learning, and memory formation. Dysregulation or aberrant	Module name: Glutamatergic Synaptic Module Module description: This module focuses on the mechanism underpinning action of glutamate, the principal excitatory neurotransmitter within the mammalian central nervous system. Glutamate mediates fast excitatory synaptic transmission through the activation of iGluRs, which function as ligand-gated ion channels. Concurrently, glutamate exerts diverse modulatory effects via mGluRs; notably, presynaptic mGluRs often serve to inhibit excessive glutamate release, providing crucial feedback regulation. The fidelity of glutamatergic signaling critically depends on the efficient clearance of synaptic glutamate, accomplished by EAATs. It predominantly located on astrocytes, thereby preventing excitotoxic neuronal damage resulting from glutamate accumulation. Collectively, iGluRs and mGluRs act in concert to fine-tune synaptic strength and plasticity. Furthermore, EAATs operate in conjunction

				modulation of this pathway is strongly associated with neurological and neurodevelopmental disorders, including autism spectrum disorder, intellectual disability, attention-deficit hyperactivity disorder, Down syndrome, Rett syndrome, and neurodegenerative diseases such as Alzheimer's and Parkinson's disease. Pathological changes in receptor activity, glutamate metabolism, and synaptic structure/function contribute significantly to these clinical outcomes. Adaptive homeostatic mechanisms, such as synaptic scaling and receptor trafficking (not detailed here), help maintain excitatory balance under physiological conditions. Confidence score: 1	with the glutamate-glutamine cycle to sustain glutamate homeostasis. Dysfunction within this intricate glutamatergic system, encompassing aberrant receptor signaling, impaired glutamate clearance, or disrupted cycling, is strongly associated with the pathophysiology of several neurological and psychiatric disorders, including epilepsy, schizophrenia, major depressive disorder, and Alzheimer's disease. Confidence score:0.88
5	Module name: Not Available Module description: Not Available Confidence score:	Module name: DNA replication initiation Module description: DNA replication initiation is a tightly regulated and evolutionarily conserved process that marks the beginning of genome	Module name: DNA Replication Module Module description: This module contains the core DNA replication machinery that duplicates the genome during S phase, including MCM helicases, primase	Module name: DNA Replication Module description: The overarching biological process is DNA replication, a fundamental mechanism ensuring accurate duplication of genetic	Module name: DNA Replication Module Module description: This module focuses on the molecular mechanism of DNA replication, encompassing three core phases: initiation, elongation, and termination.

		duplication. Across all domains of life, initiation involves a defined sequence of molecular events at specific genomic loci known as origins of replication. The process begins with the binding of origin recognition complexes or replication initiator proteins to the origin DNA. This is followed by the recruitment and loading of the replicative helicase, typically mediated by specialized helicase assembly factors. In eukaryotic cells, partial unwinding of the origin occurs at an adenine–thymine (AT)-rich region known as the DNA unwinding element (DUE), creating single-stranded DNA templates. Since DNA polymerases are unable to initiate DNA synthesis de novo, a short RNA–DNA hybrid primer is synthesized by the DNA polymerase alpha:primase complex. This primer provides the necessary 3'-OH group for DNA polymerases to begin elongation, thus enabling the establishment of the replication fork and	polymerase complexes, DNA polymerases α, δ, and ϵ, the DNA sliding clamp, and enzymes that process Okazaki fragments, ensuring high fidelity synthesis. Confidence score: 0.95	material prior to cell division. DNA replication comprises initiation, elongation, and termination. Initiation begins with origin recognition proteins binding specific DNA sequences, facilitating recruitment of DNA helicase to unwind the parental double helix and form replication forks. Single-strand binding proteins stabilize these separated strands, while topoisomerase prevents excessive supercoiling and DNA damage. Elongation involves synthesis of RNA primers by primase, allowing DNA polymerase to add nucleotides in a 5' to 3' direction. Leading strands are synthesized continuously, whereas lagging strands are synthesized discontinuously, forming Okazaki fragments. Subsequently, RNA primers are removed by DNA polymerase I (prokaryotes) or RNase H (eukaryotes), replaced by DNA nucleotides, and gaps sealed by DNA ligase, completing a continuous DNA strand. Defects or disruptions in replication fidelity are strongly linked to genomic	During replication fork establishment, DNA helicase (DnaB or MCM complex) unwinds the parental duplex DNA, facilitated by single-stranded DNA-binding proteins (SSB or RPA). Subsequently, continuous leading strand synthesis proceeds processively in the 5 to 3 direction, while lagging strand synthesis occurs discontinuously through the generation of Okazaki fragments by DNA polymerase. The process culminates in the excision of RNA primers by specialized nucleases (e.g., RNase H/FEN1) and ligation of adjacent DNA fragments by DNA ligase. The module particularly emphasizes the mechanism of replication initiation in eukaryotes, comparing the differences between unicellular eukaryotes (e.g., <i>Saccharomyces cerevisiae</i>) and multicellular eukaryotes. Unicellular eukaryotes utilize sequence-defined origins (e.g., ARS elements), whereas in metazoans, initiation site selection is governed by epigenetic determinants rather than specific DNA sequences. This was confirmed by <i>Xenopus</i> egg extract studies.
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		progression of DNA synthesis. Confidence score: 0.95		instability, contributing to cancer, congenital diseases, and premature aging disorders. Specialized mechanisms, such as DNA damage tolerance and replication restart pathways (not detailed here), help manage replication under stressful cellular conditions. Confidence score: 1	Crucially, the dysregulation of DNA replication fidelity leads to genomic instability, which is the underlying cause of oncogenesis and hereditary disorders. Confidence score:0.86
7	Module name: Electron transfer flavoprotein complex through oxidative phosphorylation Module description: The pathways involved all point to electron-transfer, and some descriptions of the involved proteins and mechanisms, which is oxidative phosphorylation. Confidence score: 0.95	Module name: Electron transport chain in oxidative phosphorylation Module description: The Electron transport chain in oxidative phosphorylation refers to the coupled processes of electron transfer and ATP synthesis occurring in the inner mitochondrial membrane. Electrons derived from metabolic substrates are passed through a series of protein complexes, namely Complex I (NADH: ubiquinone oxidoreductase), Complex II (succinate dehydrogenase), the electron transfer flavoprotein (ETF) complex, Complex III (cytochrome bc1), and Complex IV (cytochrome c oxidase), ultimately	Module name: Electron Transport Module Module description: This is a mitochondrial electron transport system centered on flavoprotein-mediated transfer in oxidative phosphorylation. The module's electron transfer flavoprotein complex links fatty acid β-oxidation to the respiratory chain via ubiquinone, coupling fatty acid metabolism to energy production. Confidence score: 0.90	Module name: Adenosine triphosphate (ATP) synthesis Module description: The overarching biological process is aerobic energy conversion, coupling the electron-transport chain (ETC) to oxidative phosphorylation, generating adenosine triphosphate (ATP). Reduced cofactors (NADH and FADH₂) generated by upstream catabolic pathways donate high-energy electrons to the ETC at Complex I or II. Electrons travel sequentially through ubiquinone (Q), Complex III, cytochrome c, and Complex IV, where molecular oxygen is reduced to water. Each redox step pumps protons across the inner mitochondrial membrane in	Module name: Redox Phosphorylation Module Module description: This module focuses on electron transfer and oxidative phosphorylation. Pathway 1 defines electron transfer activity as a directed movement of electrons between molecular entities, distinct from stochastic electron diffusion. Pathway 2 specifies the electron transfer flavoprotein (ETF) complex, a key electron carrier that shuttle electrons from acyl-CoA dehydrogenases to the ubiquinone (CoQ). Electrons subsequently traverse the inner mitochondrial membrane through the electron transport chain (ETC) complexes I-IV, culminating in the establishment of a proton gradient that drives ATP synthesis via complex V

		reducing molecular oxygen to water. These redox reactions are coupled to the translocation of protons from the mitochondrial matrix to the intermembrane space, generating an electrochemical gradient (proton motive force). This proton gradient drives ATP synthesis via Complex V (ATP synthase), which phosphorylates ADP to ATP in a process known as oxidative phosphorylation. Flavoproteins such as the ETF complex facilitate electron transfer from fatty acid-derived acyl-CoAs to the ubiquinone pool, linking β-oxidation to the respiratory chain. Together, these steps constitute a fundamental energy conversion mechanism essential for aerobic life. Confidence score: 0.95		eukaryotes (or the plasma membrane in prokaryotes), establishing an electrochemical proton-motive force. ATP synthase (Complex V) harnesses this gradient, allowing protons to flow back into the matrix/cytosol and driving the phosphorylation of ADP to ATP. The resulting ATP supplies cellular energy for virtually every energy-demanding cellular process, linking diverse catabolic pathways to biosynthesis, ion transport, signal transduction, and mechanical work. Variations such as alternative oxidases or anaerobic electron acceptors (not detailed here) modify these core principles under specialized conditions. Confidence score: 0.8	(ATP synthase). Throughout the module, the alpha and beta subunits of the ETF work together with the flavin adenine dinucleotide (FAD) cofactor to facilitate electron transfer. Complexes I-V in oxidative phosphorylation play a synergistic role in ensuring efficient energy conversion. Dysfunction in this module manifests clinically as the Inborn errors of metabolism, mitochondrial disorders and metabolic dysregulation. Confidence score:0.88
8	Module name: Adaptive immune response and regulation concerning T and B cells Module description: All pathways are directly associated with the adaptive	Module name: Module description: Adaptive immunity refers to a highly specific immune response that plays a critical role in eliminating pathogens. This system relies primarily on B and T	Module name: Adaptive Immune Receptor Signaling Module Module description: Integrated T-cell receptor and B cell receptor signaling networks for antigen-specific immune activation. It shares	Module name: Lymphocyte activation and adaptive immune signaling Module description: The module describes the overarching molecular and cellular mechanisms underlying the the antigen-specific activation of B and	Module name: Adaptive Immune Response Module Module description: This module describes the TCR and BCR signaling pathways and their essential roles in adaptive immunity. T-cell activation requires coordinated engagement of

	immune system, and T and B cell regulation and signaling, a clear call. Confidence score: 0.95	lymphocytes, which express diverse antigen-specific receptors capable of recognizing unique molecular features of foreign invaders. During infection, dendritic cells (DCs) act as peripheral sentinels that capture and process pathogens into antigenic peptides. These antigens are then presented to T cells via major histocompatibility complex (MHC) molecules. Upon recognizing a cognate antigen and receiving costimulatory signals, such as CD28 engagement with B7.1/B7.2, T cells become activated, initiating a cascade of intracellular signaling events. These cascades involve protein tyrosine kinases, phosphatases, GTP-binding proteins, and adaptor proteins, which together orchestrate T-cell proliferation, cytokine secretion, and differentiation into effector cells. B cells, which mediate humoral immunity, recognize antigens through their B cell receptor (BCR), a complex composed of membrane-bound	common kinases (e.g., SYK, LYN) and effectors (e.g., PI3K). This module also includes CD28 costimulation for T cells and enables clonal expansion and immunological memory formation. Confidence score: 0.85	T lymphocytes, the central players of adaptive immune responses. While they share several fundamental signaling mechanisms, they possess distinct pathways that define their specialized functions. cell activation is initiated by the T cell receptor (TCR) recognizing peptide-MHC complexes. Notably, optimal T cell activation requires costimulatory signals, prominently regulated by the CD28 receptor superfamily. Other costimulatory signal include ICOS (Inducible T-cell COStimulator, pathway not included), which play a key role in T follicular helper cell differentiation and germinal center responses. B cell activation is driven by antigen binding to the B cell receptor (BCR). A hallmark of B cell activation is the CD40–CD40L interaction (pathway not described) with helper T cells, which is essential for isotype class switching, somatic hypermutation, and germinal center formation. B cells also uniquely respond to BAFF (B cell activating factor) (pathway	both the TCR and costimulatory molecules such as CD28 to initiate a series of signaling cascades. And finally it leads to T cell proliferation and effector functions. Conversely, B-cell activation is triggered by BCR signaling, which activates important downstream effectors such as PTKs and PI3K, to promote antibody secretion and provide humoral immunity. In this module, TCR and CD28 work together to regulate T cell activation. And BCR signalling activates downstream signalling pathways through molecules such as BLNK and PLCγ that regulate calcium release. Defects in these pathways—such as ZAP-70 deficiency impairing TCR signaling—manifest as primary immunodeficiencies, while dysregulation contributes to autoimmune disorders like systemic lupus erythematosus. Notably, mechanistic insights have enabled transformative immunotherapies such as CAR-T cells. Confidence score:0.87
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		immunoglobulin molecules and the signaling subunits Ig-α and Ig-β. Following antigen binding, signaling is initiated through the activation of key protein tyrosine kinases, including the SRC-family kinase LYN, SYK, and the TEC-family kinase BTK. Downstream signaling molecules such as phosphatidylinositol 3-kinase (PI3K) and phospholipase C-γ2 (PLC-γ2) are activated, ultimately leading to the expression of immediate early genes. These transcriptional changes promote B cell proliferation, differentiation, antibody production, and further immune responses. Together, the TCR and BCR signaling pathways are central to the adaptive immune system, coordinating the cellular and humoral arms to achieve a robust and specific defense against pathogens. Confidence score: 0.95		not mentioned), which promotes peripheral B cell survival and maturation. Confidence score: 0.7	
9	Module name: Fatty acid biosynthetic	Module name: Fatty acid biosynthesis and TCA	Module name: Fatty Acid Biosynthesis Module	Module name: Lipid metabolism	Module name: Fatty Acid Metabolism Module

	process Module description: This module concerns the process of fatty acid synthesis, possibly through the acetyl-CoA generated through the citrate cycle (TCA cycle). Confidence score: 0.95	cycle Module description: Fatty acid biosynthesis is a fundamental anabolic process responsible for generating a wide spectrum of fatty acids, ranging from short-chain to long-chain species. These molecules serve as essential building blocks for membrane lipids, energy storage, and signaling. The biosynthesis process begins with acetyl-CoA, which is carboxylated by acetyl-CoA carboxylase (ACC) to generate malonyl-CoA. Fatty acid synthase (FASN) catalyzes the sequential addition of two-carbon units from malonyl-CoA to elongate the acyl chain, typically yielding palmitate (C16:0) as the primary product of de novo synthesis. Short-chain fatty acids (SCFAs), typically defined as fatty acids with fewer than six carbon atoms (e.g., acetate, propionate, and butyrate), can also be synthesized endogenously or produced by microbiota and contribute to host metabolism, particularly in energy regulation and	Module description: Comprehensive fatty acid synthesis system from short-chain to long-chain fatty acids. Includes de novo synthesis via acetyl-CoA carboxylase/fatty acid synthase, elongation systems, and TCA cycle integration for acetyl-CoA supply. This module also produces fatty acyl-CoAs for membranes and energy storage. Confidence score: 0.95	Module description: The overarching biological process is lipid metabolism, which covers both the catabolic and anabolic sides. Anabolic processes cover the synthesis and elongation of fatty acids of various chain lengths and the activation of fatty acids to CoA thioesters, an important step before they can be used in synthesis, degradation and incorporation into complex lipids. Catabolic processes include the TCA cycle, which provides the precursor (acetyl-CoA) for synthesis and being the endpoint of β-oxidation. Other catabolic processes beyond β-oxidation, fatty acids can be degraded through α-oxidation (for branched-chain fatty acids like phytanic acid) and ω-oxidation (a microsomal pathway that hydroxylates the terminal carbon to generate dicarboxylic acids for further oxidation). These alternative pathways (not described in this module) are particularly important when β-oxidation is impaired or when metabolizing structurally unusual fatty acids.	Module description: This module describes the processes of fatty acid biosynthesis, elongation and catabolism. Fatty acid biosynthesis initiates with the ATP-dependent carboxylation of acetyl-CoA to malonyl-CoA, catalyzed by acetyl-CoA carboxylase (ACC). Malonyl-CoA serves as the primary carbon donor for the cytosolic fatty acid synthase (FAS) complex, a multifunctional enzyme system that iteratively elongates the acyl chain through condensation, reduction, dehydration, and reduction cycles, culminating in the synthesis of saturated fatty acids, predominantly palmitate (C16:0). Subsequent elongation to very long chain fatty acids (VLCFAs) occurs on the endoplasmic reticulum membrane, mediated by membrane-bound enzyme complexes, with the elongation of very long chain fatty acids (ELOVL) family proteins conferring substrate specificity for distinct chain lengths. For catabolic utilization, long chain fatty acids are activated in the cytosol to fatty acyl-CoAs by acyl-CoA synthetases (ACSS). These activated fatty acids are
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		immune modulation. Meanwhile, long-chain and very-long-chain fatty acids are generated through further elongation and modification of palmitate via elongases and desaturases. Although fatty acid biosynthesis occurs in the cytosol, its carbon source, acetyl-CoA, is primarily derived from the tricarboxylic acid (TCA) cycle. In nutrient-replete conditions, citrate is exported from mitochondria and cleaved by ATP-citrate lyase to regenerate cytosolic acetyl-CoA, linking mitochondrial energy metabolism with lipogenesis. Fatty acids are subsequently converted to fatty acyl-CoAs by acyl-CoA synthetases for incorporation into complex lipids or further metabolism. This module highlights the integrated biochemical pathways that govern the synthesis of both short- and long-chain fatty acids and their metabolic coupling with the TCA cycle via citrate-derived acetyl-CoA. Confidence score: 0.9		Confidence score: 0.7	then translocated into the mitochondrial matrix to undergo β-oxidation. This cyclic process sequentially cleaves two-carbon units as acetyl-CoA, which subsequently enter the tricarboxylic acid (TCA) cycle for ATP generation. The dysregulation of this module is the underlying cause of pathologies including obesity, type 2 diabetes and cardiovascular disease. Confidence score:0.9
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10	Module name: MAPK-mediated apoptosis Module description: MAPK appears to be involved in apoptosis both as an activator and an inhibitor. The pathways involved in this module are all about MAPK, kinases, and apoptosis, which makes senses. Confidence score: 0.95	Module name: MAPK/ERK-mediated cell proliferation and death Module description: The ERK1/2 branch of the mitogen-activated protein kinase (MAPK) cascade is a conserved signaling module that relays extracellular cues, such as mitogens, growth factors, and GPCR ligands, from the plasma membrane to nuclear and cytoplasmic targets. Sequential activation of Raf, MEK, and ERK drives transcriptional programmes that promote cell-cycle progression, differentiation, and migration. In certain contexts, prolonged or dysregulated ERK activity instead engages apoptotic pathways by modulating key components of both the extrinsic (death-receptor) and intrinsic (mitochondrial) routes, converging on activation of effector caspases-3/7. Thus, ERK functions as a context-dependent switch linking external stimuli to either cell survival and proliferation or programmed cell death.	Module name: MAPK-Mediated Apoptosis Signaling Module Module description: This module links MAPK cascades to apoptotic pathways, with protein kinases acting as molecular switches that toggle between ERK1/2-driven survival and caspase-mediated death. The interplay between pro-survival MAPK signals and pro-apoptotic signals decides whether a cell proliferates or undergoes programmed death. Confidence score: 0.80	Module name: Cell Death, Proliferation and Differentiation Module description: The overarching biological processes outlined in this module include apoptosis (programmed cell death), proliferation, and differentiation, three connected mechanisms basic to the determination of cell fate in the development of the organism, maintenance of the tissue, and development of disease. Apoptosis pathways emphasize the regulated balance between survival stimuli (e.g., NGF, IL-3) and death signals (extrinsic/intrinsic triggers), determining whether cells persist or undergo programmed elimination. Proliferation and differentiation of the cell are governed principally by the MAPK signaling cascades, in particular via ERK1/2, in response to extracellular stimuli (e.g., growth factors, cytokines, and mitogens). These signaling networks integrate diverse environmental signals to resolve the cellular outcomes. Main molecular mediators include	Module name: Apoptosis and MAPK Signaling Pathway Module Module description: This module integrates main mechanisms of apoptosis and MAPK signaling pathways. Apoptosis, a genetically programmed cell death process, is primarily initiated via three distinct pathways: the extrinsic, intrinsic and granzyme B pathways. These pathways converge to activate effector caspases, notably caspase-3 and caspase-7, which execute the terminal stages of apoptosis by cleaving key cellular substrates. The module also describes the MAPK signaling cascade, a critical regulator of diverse cellular responses. This cascade transduces signals through major branches involving ERK1/2, JNK, p38 proteins and ERK5. The module emphasizes the importance of the critical balance between pro-apoptotic and anti-apoptotic signals that determine cell fate. For instance, TNF family can paradoxically initiate both anti-apoptotic (e.g., via NF-κB activation) and pro-apoptotic signaling cascades. Furthermore, the module highlights the context-
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		Confidence score: 0.8		protein kinases, which transduce and amplify signals to orchestrate these essential cellular processes. Dysfunction or misregulation within these integrated networks underlies numerous pathological states, including cancer, developmental abnormalities, and degenerative diseases. Adaptive regulatory mechanisms such as feedback loops, and cross-talk (not detailed here) then fine-tune these pathways to maintain cellular homeostasis and normal tissue function. Confidence score: 0.6	dependent roles of specific MAPK pathways in apoptosis regulation: while the ERK1/2 cascade predominantly promotes cell proliferation and survival, both the JNK and p38 MAPK pathways can potentiate apoptosis under stress conditions. Dysregulation of apoptosis is a well-established hallmark closely associated with the pathogenesis of cancer, autoimmune disorders and neurodegenerative diseases. Similarly, aberrant activation or inhibition within the MAPK signaling network is implicated in numerous inflammatory conditions and tumorigenesis. Confidence score: 0.85
11	Module name: Muscel cell differentiation Module description: All pathways appear to be either about regulation or negative regulation of muscle cell differentiation, so this module should be about muscle cell differentiation. Confidence score: 0.95	Module name: Putative negative regulation of muscle cell differentiation Module description: A process that most likely inhibits or attenuates the differentiation of relatively unspecialized cells into specialized muscle cells, particularly striated muscle cells such as those found in cardiac and skeletal muscle. Striated muscle is characterized by transverse banding patterns and includes voluntary and	Module name: Striated Muscle Differentiation Module Module description: This module is a regulatory network controlling striated muscle cell differentiation through coordinated positive and negative regulation. All pathways converge specifically on cardiac and skeletal muscle development, functioning as a molecular regulator to control differentiation timing and extent.	Module name: Myogenic Differentiation & Regulation Module description: The overarching biological process described by this module is the differentiation of progenitor cells (myoblasts) into mature muscle fibers (myotubes), focusing specifically on striated muscle of skeletal and cardiac origin. This process is regulated by myogenic regulatory factors (MRFs), including MyoD,	Module name: Regulation of Muscle Cell Differentiation Module Module description: This module focuses on the processes of positive and negative regulation of muscle cell differentiation, particularly differentiation into striated muscle cell. It details the specialization of precursor cells into rhabdomyoblasts and the underlying regulatory networks. A critical aspect of striated muscle differentiation

		cardiac muscle types. The negative regulation may reduce the frequency, rate, or extent of this differentiation process. Confidence score:0.9	Confidence score: 0.95	Myf5, myogenin, and MRF4. Their actions on muscle-specific genes lead to cell cycle exit, fusion, and formation of the contractile apparatus. Myogenic differentiation is a carefully regulated balance of positive and negative signals. Pro-myogenic signals promote differentiation, while inhibitory pathways, such as those involving Myostatin or Notch signaling, act as brakes to prevent premature or excessive muscle formation. This balance, when dysregulated, is implicated in a number of diseases, including muscular dystrophies, rhabdomyosarcoma, and sarcopenia. Confidence score: 1	involves extensive cytoskeletal reorganization, profound metabolic reprogramming closely linked to mitochondrial bioenergetics and dynamics, and the establishment of the contractile apparatus. Positive regulatory mechanisms facilitate the acquisition of myogenic characteristics by undifferentiated progenitor cells, driving them towards the mature striated phenotype. Conversely, negative regulatory mechanisms act to constrain or attenuate this differentiation program. The precise coordination of myogenesis is critically dependent on a network of key transcription factors (such as MyoD ,MRF4) and the integration of specific signaling pathways (notably the Wnt and the Notch pathway). Dysregulation of these finely balanced regulatory processes is etiologically linked to significant pathological conditions, including myotonic dystrophy and various forms of cardiomyopathy. Confidence score:0.90
12	Module name:	Module name: Proteasomal	Module name: Ubiquitin-	Module name: Protein	Module name: Ubiquitin-

	Ubiquitination-mediated proteolysis Module description: All pathways are concerning different processes of the ubiquitination-mediate proteolysis through proteosome, this is a clear call. Confidence score: 0.95	degradation via ubiquitination Module description: Proteasomal degradation via ubiquitination is a major pathway for regulated protein turnover in eukaryotic cells. In this process, proteins destined for degradation are covalently tagged with ubiquitin, a highly conserved 76-amino acid protein, through the coordinated action of a three-enzyme cascade: E1 (ubiquitin-activating enzyme), E2 (ubiquitin-conjugating enzyme), and E3 (ubiquitin ligase). E3 ligases confer substrate specificity by recognizing and targeting specific proteins for ubiquitination. Polyubiquitinated proteins are subsequently recognized and degraded by the proteasome (the most common form is 26S), thereby maintaining proteostasis and regulating diverse cellular processes. Confidence score: 0.95	Proteasome Degradation Module Module description: The complete ubiquitin-proteasome system for targeted protein degradation. This module encompasses E1-E2-E3 ubiquitin ligase cascades and 26S proteasome machinery. Specifically, it includes K48-linked polyubiquitin chains target proteins for degradation, regulating cell cycle, protein quality control, and stress responses. Confidence score: 0.85	degradation by the Ubiquitin-Proteasome System Module description: This module describes the Ubiquitin-Proteasome System (UPS), which is essential for the targeted degradation of most short-lived proteins within eukaryotic cells. The UPS maintains cellular homeostasis by precisely controlling the levels of key regulatory proteins, including cell-cycle cyclins and transcription factors, and eliminating misfolded or damaged proteins. Target protein degradation occurs through ubiquitination, a multistep enzymatic process involving three distinct classes of enzymes. Initially, a ubiquitin-activating enzyme (E1) forms a thioester bond with ubiquitin, activating it. In humans, there are over 15 different E1 enzymes. Activated ubiquitin is subsequently transferred to a ubiquitin-conjugating enzyme (E2), of which approximately 30 variants exist, each capable of partnering with multiple E3 ligases. Finally, an E3	mediated Protein Degradation Module Module description: This module describes the process of ubiquitination and their key role in protein degradation. Ubiquitination attaches ubiquitin to target proteins through the E1-E2-E3 enzymatic cascade reaction, forming monoubiquitinated or polyubiquitinated chains that further perform various functions. The specific topology of the ubiquitin modification dictates the fate of the substrate protein. Notably, polyubiquitin chains linked through lysine 48 (K48) predominantly target substrates for degradation by the 26S proteasome. In contrast, chains linked through lysine 63 (K63) are primarily associated with non-proteolytic functions, including signal transduction pathways and DNA damage repair. E3 ubiquitin ligases confer substrate specificity to the entire ubiquitination machinery, thereby playing a pivotal regulatory role. Major classes of E3 ligases, such as CRLs and APC/C, are fundamental regulators of processes including cell cycle progression and immune responses. These complexes
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				ubiquitin ligase facilitates the transfer of ubiquitin to a specific lysine residue on the target protein through the formation of an isopeptide bond. E3 ligases, the most diverse enzyme class within the UPS, confer substrate specificity, determining precisely which proteins are targeted for degradation. Substrate recognition by E3 ligases often relies on specific degradation signals or degrons within target proteins, which can be exposed by phosphorylation or other post-translational modifications. Polyubiquitinated proteins are recognized and processed by the 26S proteasome, a complex responsible for unfolding substrates, removing ubiquitin tags for recycling, and ultimately degrading the proteins into small peptides. The UPS is dynamically regulated by deubiquitinating enzymes (DUBs), which can reverse ubiquitination and modulate protein stability. Given its critical role in cellular regulation, dysregulation of the UPS has been implicated in	collaborate with substrate recognition subunits to ensure ubiquitination specificity. Ubiquitination and deubiquitinating enzymes work together to maintain protein homeostasis. Dysregulation of the ubiquitination is strongly implicated in the pathogenesis of various diseases, including cancer, neurodegenerative disorders, and immune dysfunctions. Confidence score:0.84
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				numerous human diseases, including cancer, neurodegenerative disorders such as Alzheimer's and Parkinson's diseases, and immune dysfunction. Consequently, E3 ligases and the proteasome represent significant targets for therapeutic intervention. Confidence score: 1	
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Supplementary Note 1. Key limitations of existing pathway similarity metrics

Multiple algorithms exist for computing pathway similarity, and they can be broadly classified into two categories. First, component-based similarity metrics, such as the Jaccard index³, overlap coefficient⁴, Dice Coefficient⁵, and Cohen's Kappa⁶, quantify the degree of overlap in molecular components (e.g., genes, proteins, or metabolites) between pathway pairs. These methods are simple, interpretable, and applicable across different databases. However, they suffer from key limitations: they are highly sensitive to pathway size, often inflate similarity for large pathways, and disregard descriptive or contextual information⁷.

Second, semantic similarity metrics, including the Wang⁸, Resnik⁹, and Lin¹⁰ algorithms, leverage the structured hierarchy of ontologies such as the GO^{11,12}. These algorithms can capture nuanced parent-child relationships and broader biological context¹³. However, they are restricted to curated ontologies like GO and cannot be applied to general pathway resources such as KEGG and Reactome¹⁴. Furthermore, their dependence on static ontology graphs limits their adaptability to emerging biological knowledge and cannot be used across databases⁸.

Third, text-based similarity methods, such as PAVER¹⁵, use text embeddings to represent pathways and compute pathway similarity. For example, while PAVER can capture semantic relationships and work across different databases, it has notable drawbacks. It relies on precomputed embeddings that may not include updates from the latest pathway databases¹⁵. Users can utilize the built-in precomputed embeddings, but these may be outdated. Alternatively, users must compute embeddings themselves, which requires technical expertise. This dependency on static embeddings and the complexity of generating updated embeddings limit the practical applicability of PAVER.

c. Existing pathway similarity metrics

Component-Based Similarity. These similarity metrics quantify the overlap between pathway/term components and enable cross-database similarity calculations¹⁶.

1. Kappa coefficient. The Kappa coefficient is to quantify the degree of overlapping between pathways beyond what would be expected by random chance.

Given two pathways A and B with n total molecules annotated to them, construct a 2×2 contingency table:

	In pathway B	Not in pathway B	Row total
In pathway A	a	b	a + b
Not in pathway A	c	d	c + d
Column total	a + c	b + d	a + b + c + d

Where a, b, c, and d represent the counts for molecules, and $n = a + b + c + d$.

The observed agreement probability is: $p_{obs} = \frac{a + d}{n}$

The expected random agreement probability under independence assumption is:

$$p_{exp} = \frac{(a + b)(a + c)}{n^2} + \frac{(c + d)(b + d)}{n^2}$$

The final kappa coefficient is defined as: $K(A, B) = \frac{p_{obs} - p_{exp}}{1 - p_{exp}}$

Kappa coefficient ranges from -1 to 1, where: 1 indicates perfect agreement between the two pathways' component sets. 0 indicates the observed agreement is exactly what would be expected

by chance. There is no real agreement, only random overlap. < 0 indicates the observed agreement is worse than what would be expected by chance. This suggests a systematic disagreement between the component sets.

2. Dice coefficient. It is calculated by doubling the number of shared components and dividing by the total number of components in both pathways. The Dice coefficient is defined as:

$$D(A, B) = \frac{2 * |A \cap B|}{|A| + |B|},$$

where A and B are the sets of molecules annotated to each pathway. This metric ranges from 0 (no common components) to 1 (identical component sets). The Dice coefficient is less sensitive to differences in set sizes than the Jaccard index but can still be influenced when there is a very large discrepancy between the number of components in A and B.

Semantic Similarity. These metrics quantify semantic similarities between GO terms within the same subontology. The semantic similarity metrics employed in this study include the Wang method (based on directed acyclic graph structure) and information content-based methods, as described below^{13,17}:

In IC-based methods, each GO term t is assigned an information content $IC(t) = -\log p(t)$, where $p(t)$ is the annotation-based probability of observing term t in the GO corpus (e.g., $p(t) = \frac{k}{N}$ with k genes annotated to t and N the total genes annotated in GO). Because GO allows multiple parents, two terms (term a and term b) can share several ancestors; the key ancestor used by IC methods is the most informative common ancestor (MICA, term c), which is the common ancestor with the highest IC. All four methods below differ mainly in how they utilize $IC(a)$, $IC(b)$ and $IC(c)$.

1. The Resnik method is defined as:

$$sim(a, b) = IC(c).$$

It rewards pairs whose closest shared ancestor is very specific (high IC).

2. The Lin method normalizes the Resnik's similarity by the specificity of the query terms themselves,

$$sim(a, b) = \frac{IC(c)}{(IC(a) + IC(b)) / 2}.$$

This ratio ranges from 0 to 1, and is high only when both terms are close to their MICA and have comparable specificity.

3. The Rel method refines Lin by penalizing MICAs that are common in the corpus. It multiplies Lin's similarity by a specificity factor $1 - p(c)$,

$$sim_{Rel}(a, b) = sim_{Lin}(a, b) * [1 - p(c)].$$

The effect is to keep similarities high only when the two terms share an informative and relatively rare ancestor; if the MICA is frequent, the factor shrinks the similarity accordingly.

4. The Jiang method calculates similarity from the perspective of semantic distance. It first defines a distance metric $D(a, b)$ based on the IC of the two terms and their MICA: $D(a, b) = IC(a) + IC(b) - 2IC(c)$. This distance value represents the amount of information required to transform one term into the other. The final similarity score is then derived as an inverse function of this distance,

$$sim(a, b) = 1 - \min(1, D(a, b)).$$

This metric ranges from [0,1], where a smaller distance corresponds to higher similarity.

Supplementary Note 3. Biological connections among GO:0098985, GO:0098978, and hsa04724.

These three pathway identifiers represent closely related aspects of glutamatergic neurotransmission. GO:0098985 (asymmetric, glutamatergic, excitatory synapse) is a specialized subtype of GO:0098978 (glutamatergic synapse), both describing cellular components in the Gene Ontology. The KEGG pathway hsa04724 (glutamatergic synapse) describes the biological processes that occur at these synaptic structures. Despite originating from different databases, all three terms describe complementary aspects of the same biological process, glutamatergic synaptic transmission. This biological coherence underlies the high similarity scores produced by biotext embedding (0.64–0.74) and demonstrates the method’s ability to capture functional relationships across database boundaries.

Supplementary Note 4. Explanation for differences between ground truth and clustering result generated by louvain algorithm (similarity cutoff: 0.55)**Difference1: The hsa04210 (Apoptosis) pathway failed to cluster with pathways in the expected Functional_module_10.**

In the curated dataset, hsa04210 (Apoptosis) was expected to cluster with pathways in the MAPK signaling module. This is because hsa04010 (MAPK signaling pathway) and hsa04210 are documented in the KEGG database as related pathways, since MAPK signaling is a key upstream regulator of apoptosis. Typical members of MAPKs, such as C-Jun N-terminal kinase (JNK), p38 MAPK, and ERK1/2, can mediate both pro-apoptotic and anti-apoptotic processes, balancing cell survival and death¹⁸. However, our biotext embedding generated only moderate similarities (0.33–0.52) between hsa04210 and other pathways in this module. When we applied a similarity cutoff of 0.55, hsa04210 became a singleton, separating it from the other pathways. These moderate similarities occurred because the pathway descriptions provided by the databases hardly refer to the association between apoptosis and MAPK, instead putting emphasis on cell survival-related processes. This discrepancy demonstrates that MAPA’s similarity threshold provides users with analytical flexibility. At a higher cutoff (e.g., 0.55), we obtain specific functional modules focused on survival, while lower thresholds capture broader functional themes encompassing general cell fate control. This allows researchers to tailor their analysis to match their specific research questions.

Difference2: hsa00190 (Oxidative phosphorylation) failed to cluster with its expected Functional_module_7.

The hsa00190 pathway was expected to cluster with GO:0009055 (electron transfer activity) and GO:0045251 (electron transfer flavoprotein complex). These processes are highly related, as they are functionally interdependent parts of cellular energy production. Specifically, electron transfer activity and the ETF complex directly power oxidative phosphorylation by moving electrons through the respiratory chain to create the proton gradient for ATP synthesis¹⁹.

Despite this functional link, hsa00190 failed to cluster with the two GO terms because it only achieved a moderate similarity score of approximately 0.4 with each. We hypothesize this low similarity is due to the brevity of the pathway’s description (Oxidative phosphorylation), which provides insufficient semantic context for the text embedding model. This hypothesis is supported by observing other terms in the dataset with similarly moderate scores (~0.43). These terms typically fall into two categories: One is those with very short descriptions (e.g., hsa00062, “Fatty acid elongation”). The other one is those that share superficial keywords like “phosphorylation,” leading to vocabulary overlap without a highly functional connection (e.g., GO:0004672 (protein kinase activity)). These results suggest that the performance of the

biotext embedding similarity might be influenced by text length and simple vocabulary overlap when calculating similarities between biological concepts.

Supplementary Note 5. R version and R packages used for data analysis in this study

R version 4.5.0 (2025-04-11)

Platform: aarch64-apple-darwin20

Running under: macOS Sequoia 15.6

attached base packages:

[1] stats graphics grDevices utils datasets methods base

other attached packages:

[1] paletteer_1.6.0 ggthemes_5.1.0 AnnotationHub_3.16.0 BiocFileCache_2.16.0 dbplyr_2.5.0
BiocGenerics_0.54.0
[7] generics_0.1.4 data.table_1.17.4 rio_1.2.3 lubridate_1.9.4 forcats_1.0.0 stringr_1.5.1
[13] purrr_1.0.4 readr_2.1.5 tidyr_1.3.1 tibble_3.2.1 tidyverse_2.0.0 mapa_2.1.0
[19] ggplot2_3.5.2 dplyr_1.1.4 masstools_1.0.15 massdataset_1.0.34 magrittr_2.0.3
r4projects_0.1.6

loaded via a namespace (and not attached):

[1] IRanges_2.42.0 R.methodsS3_1.8.2 dichromat_2.0-0.1 koRpus_0.13-8
[5] progress_1.2.3 vsn_3.76.0 Biostrings_2.76.0 vctrs_0.6.5
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Supplementary Note 6. Prompt engineering

Prompt design for literature reranking

System Message:

You are an AI tasked with identifying the most relevant and valuable articles for the given module.

User Message Template (example fields shown as placeholders):

The module is defined by the following pathways: **{pathways}**.

The module also focuses on the following molecules: **{molecules}**.

Below is an abstract or chunk of text. Please:

Rank its relevance to the module in terms of how well it aligns with the pathways and molecules.

Remove unrelated information such as author names, affiliations, and non-relevant metadata.

Return the cleaned text and assign it a relevance score (from 0 to 1, where 1 is highly relevant).

{chunk}

Please respond in the following JSON format:

```
{
  "relevance_score": <score>,
  "cleaned_text": "<cleaned text>"
}
```

Prompt design for Functional module annotation (example fields shown as placeholders)

****Task Description****

You are an AI tasked with generating a ****biological module name, a research summary, and a confidence score**** based on the following information.

****Your Task****

1. ****Identify the core biological process**** that describes this module.
2. ****Analyze functional relationships**** among **{query_molecules}**, considering:
 - Shared pathways or mechanisms
 - Synergistic/antagonistic interactions
 - Disease relevance
3. ****Avoid listing isolated {query_product} functions****; instead, group **{query_product}** with similar functions and analyze their collective roles.
4. ****Name the module concisely****, ensuring it highlights a key biological process, disease association, or molecular mechanism rather than a generic term.
5. Assign a ****confidence score (0.00 - 1.00)****. This score reflects how strongly the input **{query_molecules}** and pathways overlap or converge on a single, coherent biological process. For instance, if the pathways are highly related (e.g., serine-type endopeptidase inhibitor activity and endopeptidase inhibitor activity), the score should be high (close to 1.00). If the pathways are largely unrelated (e.g., negative regulation of inflammatory response and camera-type eye development), the score should be low (close to 0.00). Consider whether these pathways genuinely point to one biological theme, whether the **{query_molecules}** are meaningfully involved in those pathways, and whether literature or research supports a functional connection.
6. ****Incorporate research insights****, ensuring the generated summary reflects findings from the provided literature (if provided).

****Example Input****

The module is defined by the following pathways:

Transmembrane receptor protein kinase activity

(Catalysis of the reaction: ATP + a protein tyrosine = ADP + protein tyrosine phosphate.)

Transmembrane receptor protein tyrosine kinase activity

(Combining with a signal and transmitting the signal from one side of the membrane to the other to initiate a change in cell activity.)

Protein tyrosine kinase activity

(Catalysis of the reaction: a protein + ATP = a phosphoprotein + ADP.)

The module also focuses on the following genes and/or proteins:

transforming growth factor beta receptor 2, insulin like growth factor 1 receptor, EPH receptor A2, epidermal growth factor receptor, platelet derived growth factor receptor alpha, erb-b2 receptor tyrosine kinase 2, ret proto-oncogene, fms related receptor tyrosine kinase 4.

Below are related articles:

Title: Inhibition of DDR2 reveals kinase-dependent and kinase-independent functions.

Text: Tyrosine kinase signaling plays a crucial role in fibrosis. DDR2, a receptor tyrosine kinase, regulates fibroblast apoptosis and fibrosis progression. Inhibitors like WRG-28 block DDR2 activation, reducing fibrosis severity.

Title: Oncogene-driven receptor tyrosine kinase expression in cancer.

Text: Overexpression of receptor tyrosine kinases like DDR2, RET, and PDGFRA correlates with poor survival in neuroblastoma. The multikinase inhibitor sitravatinib effectively targets DDR2, RET, and PDGFRA, reducing tumor proliferation and metastasis.

Example Output

```
{
  "module_name": "Receptor Tyrosine Kinase Signaling in Fibrosis and Cancer",
  "summary": "This module focuses on the role of receptor tyrosine kinases (RTKs) in fibrosis and cancer, particularly emphasizing pathways related to transmembrane receptor protein kinase activity. Key genes involved include DDR2, PDGFRA, RET, EGFR, and EPH receptors, which regulate cellular signaling in fibroblasts, neuroblastoma, and other disease contexts. The studies highlight DDR2's central role in fibrosis through collagen autocrine signaling and its involvement in cancer progression. Inhibitors like sitravatinib and WRG-28 have shown promise in targeting DDR2 and other RTKs, affecting tumor growth and fibrosis progression. Additionally, DDR2-mediated PIK3C2α signaling has been implicated in fibroblast function and fibrosis, presenting a potential therapeutic target."
  "confidence_score": "0.95"
}
```

Actual Input for Generation

The biological module is defined by following pathways: {**pathway_info**}

The module also focuses on following {**query_molecules**}: {**query_molecule_names**}

Below are related articles: {**combined_texts**}

Final Output

Please provide your response in **JSON format**, strictly following this structure:

```
{
  "module_name": "Your concise module name here",
  "summary": "A detailed, literature-integrated explanation of the module's function, key {query_molecules}, and relevant biological processes.",
}
```

"confidence_score": "A value between 0.00 and 1.00, based on {query_molecules}-pathway association and overall functional coherence."
}

When generating your answer, pay attention to the following:

module_name should be concise and descriptive, highlighting the core biological process or disease relevance.

summary should include:

The central biological function or process.

Key {query_molecules} and their roles within the identified pathways.

Insights from the provided articles or research (if provided).

confidence_score should reflect how well the {query_molecules} and pathways associated with each other and support a coherent biological process, with 1.00 indicating very high confidence and 0.00 indicating minimal or no overlap.

Prompt design for meta-module annotation

Task Description

You will be given a meta-module containing multiple functional modules from different tissues. Your goal is to:

1. ****Generate a concise, biologically meaningful meta-module name**** (3-8 words)
2. ****Provide a comprehensive summary**** explaining the biological significance of this meta-module.

Input Information Format

For each functional module in the meta-module, you will receive:

- ****Module Summary****: Functional description of the module
- ****Tissue****: The tissue where this module was identified

Analysis Guidelines

For Meta-module Naming:

- Focus on the ****core biological process**** or ****pathway theme**** shared across functional modules
- Consider ****tissue specificity**** if relevant (e.g., "Cardiac-specific", "Multi-tissue")
- Use standard biological terminology
- Keep it concise but informative

For Summary Generation:

- ****Identify common biological themes**** across all modules in the meta-module
- ****Mention key pathways or processes**** represented in the meta-module
- ****Consider cross-tissue communication**** or shared regulatory mechanisms
- ****Highlight their biological relevance****

Output Format

Respond with a valid JSON object using exactly this format (no markdown, no code blocks, no additional text):

```
{"meta-module_name": "Your concise meta-module name here", "summary": "A comprehensive biological summary explaining the meta-module's significance, tissue patterns, and potential biological implications"}
```

Start your response directly with the opening brace { and end with the closing brace }.

Actual Input for Generation

****Functional modules information in this meta-modules**:**

{module_data}

Example Analysis Approach

1. Read through all functional module summaries to identify common themes
2. Note tissue distribution
3. Consider biological relationships between tissues
4. Formulate a name that captures the essence of the meta-module
5. Write a summary that explains the biological story this meta-module tells

Please analyze the provided meta-modules and generate the meta-module name and summary following the guidelines above.

Supplementary Note 7. Real functional module set

Module 4

Pathway 1: GO:0098978 (GO) glutamatergic synapse

A synapse that uses glutamate as a neurotransmitter.

Pathway 2: GO:0098985 (GO) asymmetric, glutamatergic, excitatory synapse

A neuron-to-neuron synapse with a postsynaptic density, that uses glutamate as a neurotransmitter and whose activity results in excitatory postsynaptic potentials.

Pathway 3: hsa04724 (KEGG) Glutamatergic synapse

Glutamate is the major excitatory neurotransmitter in the mammalian central nervous system(CNS). Glutamate is packaged into synaptic vesicles in the presynaptic terminal. Once released into the synaptic cleft, glutamate acts on postsynaptic ionotropic glutamate receptors (iGluRs) to mediate fast excitatory synaptic transmission. Glutamate can also act on metabotropic glutamate receptors (mGluRs) and exert a variety of modulatory effects through their coupling to G proteins and the subsequent recruitment of second messenger systems. Presynaptically localized Group II and Group III mGluRs are thought to represent the classical inhibitory autoreceptor mechanism that suppresses excess glutamate release. After its action on these receptors, glutamate can be removed from the synaptic cleft by EAATs located either on the presynaptic terminal, neighboring glial cells, or the postsynaptic neuron. In glia, glutamate is converted to glutamine, which is then transported back to the presynaptic terminal and converted back to glutamate.

Module 5

Pathway 1: hsa03030 (KEGG) DNA replication

A complex network of interacting proteins and enzymes is required for DNA replication. Generally, DNA replication follows a multistep enzymatic pathway. At the DNA replication fork, a DNA helicase (DnaB or MCM complex) precedes the DNA synthetic machinery and unwinds the duplex parental DNA in cooperation with the SSB or RPA. On the leading strand, replication occurs continuously in a 5 to 3 direction, whereas on the lagging strand, DNA replication occurs discontinuously by synthesis and joining of short Okazaki fragments. In prokaryotes, the leading strand replication apparatus consists of a DNA polymerase (pol III core), a sliding clamp (beta), and a clamp loader (gamma delta complex). The DNA primase (DnaG) is needed to form RNA primers. Normally, during replication of the lagging-strand DNA template, an RNA primer is removed either by an RNase H or by the 5 to 3 exonuclease activity of DNA pol I, and the DNA ligase joins the Okazaki fragments. In eukaryotes, three DNA polymerases (alpha, delta, and epsilon) have been identified. DNA primase forms a permanent complex with DNA polymerase alpha. PCNA and RFC function as a clamp and a clamp loader. FEN 1 and RNase H1 remove the RNA from the Okazaki fragments and DNA ligase I joins the DNA.

Pathway 2: R-HSA-68952 (Reactome) DNA replication initiation

DNA polymerases are not capable of de novo DNA synthesis and require synthesis of a primer, usually by a DNA-dependent RNA polymerase (primase) to begin DNA synthesis. In eukaryotic cells, the primer is synthesized by DNA polymerase alpha:primase. First, the DNA primase portion of this complex synthesizes approximately 6-10 nucleotides of RNA primer and then the DNA polymerase portion synthesizes an additional 20 nucleotides of DNA (Frick & Richardson 2002; Wang et al 1984).

Pathway 3: R-HSA-69306 (Reactome) DNA Replication

Studies in the past decade have suggested that the basic mechanism of DNA replication initiation is conserved in all kingdoms of life. Initiation in unicellular eukaryotes, in particular *Saccharomyces cerevisiae* (budding yeast), is well understood, and has served as a model for studies of DNA replication initiation in multicellular eukaryotes, including humans. In general terms, the first step of initiation is the binding of the replication initiator to the origin of replication. The replicative helicase is then assembled onto the origin, usually by a helicase assembly factor. Either shortly before or shortly after helicase assembly, some local unwinding of the origin of replication occurs in a region rich in adenine and thymine bases (often termed a DNA unwinding element, DUE). The unwound region provides the substrate for primer synthesis and initiation of DNA replication. The best-defined eukaryotic origins are those of *S. cerevisiae*, which have well-conserved sequence elements for initiator binding, DNA unwinding and binding of accessory proteins. In multicellular eukaryotes, unlike *S. cerevisiae*, these loci appear not to be defined by the presence of a DNA sequence motif. Indeed, choice of replication origins in a multicellular eukaryote may vary with developmental stage and tissue type. In cell-free models of metazoan DNA replication, such as the one provided by *Xenopus* egg extracts, there are only limited DNA sequence specificity requirements for replication initiation (Kelly & Brown 2000; Bell & Dutta 2002; Marahrens & Stillman 1992; Cimborra & Groudine 2001; Mahbubani et al 1992; Hyrien & Mechali 1993).

Module 7

Pathway 1: GO:0009055 (GO) electron transfer activity

A molecular function representing the directed movement of electrons from one molecular entity to another, typically mediated by electron carriers or acceptors, resulting in the transfer of energy and/or the reduction-

oxidation (redox) transformation of chemical species. This activity is fundamental to various biological processes, including cellular respiration and photosynthesis, as well as numerous enzymatic reactions involved in metabolic pathways.

Pathway 2: GO:0045251 (GO) electron transfer flavoprotein complex

A protein complex facilitating the electron transfer from an acyl-CoA molecule to ubiquinone via its flavin adenine dinucleotide (FAD) cofactor. Usually contains an alpha and a beta subunit and the structural cofactor adenosine monophosphate (AMP). Part of a system that oxidizes an acyl-CoA molecule and reduces ubiquinone and other acceptors in the electron transport system.

Pathway 3: hsa00190 (KEGG) Oxidative phosphorylation

NA

Module 8

Pathway 1: GO:0002250 (GO) adaptive immune response

An immune response mediated by cells expressing specific receptors for antigens produced through a somatic diversification process, and allowing for an enhanced secondary response to subsequent exposures to the same antigen (immunological memory).

Pathway 2: GO:0006955 (GO) immune response

Any immune system process that functions in the calibrated response of an organism to a potential internal or invasive threat.

Pathway 3: hsa04660 (KEGG) T cell receptor signaling pathway

Activation of T lymphocytes is a key event for an efficient response of the immune system. It requires the involvement of the T-cell receptor (TCR) as well as costimulatory molecules such as CD28. Engagement of these receptors through the interaction with a foreign antigen associated with major histocompatibility complex molecules and CD28 counter-receptors B7.1/B7.2, respectively, results in a series of signaling cascades. These cascades comprise an array of protein-tyrosine kinases, phosphatases, GTP-binding proteins and adaptor proteins that regulate generic and specialised functions, leading to T-cell proliferation, cytokine production and differentiation into effector cells.

Pathway 4: hsa04662 (KEGG) B cell receptor signaling pathway

B cells are an important component of adaptive immunity. They produce and secrete millions of different antibody molecules, each of which recognizes a different (foreign) antigen. The B cell receptor (BCR) is an integral membrane protein complex that is composed of two immunoglobulin (Ig) heavy chains, two Ig light chains and two heterodimers of Ig-alpha and Ig-beta. After BCR ligation by antigen, three main protein tyrosine kinases (PTKs) -the SRC-family kinase LYN, SYK and the TEC-family kinase BTK- are activated. Phosphatidylinositol 3-kinase (PI3K) and phospholipase C-gamma 2 (PLC-gamma 2) are important downstream effectors of BCR signalling. This signalling ultimately results in the expression of immediate early genes that further activate the expression of other genes involved in B cell proliferation, differentiation and Ig production as well as other processes.

Pathway 5: R-HSA-1280218 (Reactome) Adaptive Immune System

Adaptive immunity refers to antigen-specific immune response efficiently involved in clearing the pathogens. The adaptive immune system is comprised of B and T lymphocytes that express receptors with remarkable diversity tailored to recognize aspects of particular pathogens or antigens. During infection, dendritic cells (DC) which act as sentinels in the peripheral tissues recognize and pick up the pathogen in the form of antigenic determinants and then process these antigens and present them to T cells. These T cells of appropriate specificity respond to the antigen, and either kill the pathogen directly or secrete cytokines that will stimulate B lymphocyte response. B cells provide humoral immunity by secreting antibodies specific for the pathogen or antigen.

Pathway 6: R-HSA-388841 (Reactome) Regulation of T cell activation by CD28 family

Optimal activation of T lymphocytes requires a carefully orchestrated interplay between two key signals. The first signal originates from the T cell receptor (TCR) recognizing a specific antigen. However, this initial encounter is insufficient on its own. A crucial “costimulatory” signal is needed for full T cell activation, delivered by the engagement of costimulatory receptors, such as CD28, on the T cell surface with their corresponding ligands on antigen-presenting cells (APCs) (Chen & Flies 2013, Acuto et al. 2003, Slavik et al. 1999). The CD28 superfamily is central to costimulation, encompassing a diverse group of receptors including CD28, CTLA-4, ICOS, PD-1, and BTLA. These receptors have both positive and negative influences on T cell activation. CD28 and ICOS provide a positive boost, while CTLA-4, PD-1, and BTLA act as inhibitory brakes. CD28 and CTLA-4 bind to B7 family ligands CD80 (B7.1) and CD86 (B7.2), with CD28 delivering stimulatory signals and CTLA-4 providing inhibitory signals. ICOS interacts with ICOS ligand (ICOS-L), enhancing T cell activation and function. PD-1 binds to PD-L1 and PD-L2, mediating inhibitory signals that modulate the immune response. BTLA engages with HVEM (Herpesvirus entry mediator), transmitting inhibitory signals. This balance between stimulatory and inhibitory signals is essential, allowing the immune system to mount effective responses against pathogens while preventing the induction of T cell unresponsiveness and apoptosis (Chen & Flies 2013, Sharpe & Freeman 2002, Carreno & Collins 2002). The expression of these receptors varies: CD28 is constantly present on naive T cells, whereas CTLA-4 expression depends on prior CD28 engagement. ICOS, PD-1, and BTLA are induced only after initial T cell stimulation. These variations in expression ensure precise regulation of T cell responses at different stages of activation (Sharpe & Freeman 2002). The receptors themselves also exhibit structural differences. CD28, CTLA-4, and ICOS share a single extracellular domain resembling an immunoglobulin molecule (IgV-like). In contrast, BTLA has a distinct structure with an IgC-like domain. Similarly, the costimulatory ligands, such as B7-1 and B7-2, have unique structural features that enable specific interactions with their partner receptors

Pathway 7: R-HSA-983705 (Reactome) Signaling by the B Cell Receptor (BCR)

Mature B cells express IgM and IgD immunoglobulins which are complexed at the plasma membrane with Ig-alpha (CD79A, MB-1) and Ig-beta (CD79B, B29) to form the B cell receptor (BCR) (Fu et al. 1974, Fu et al. 1975, Kunkel et al. 1975, Van Noesel et al. 1992, Sanchez et al. 1993, reviewed in Brezski and Monroe 2008). Binding of antigen to the immunoglobulin activates phosphorylation of immunoreceptor tyrosine-based activation motifs (ITAMs) in the cytoplasmic tails of Ig-alpha and Ig-beta by Src family tyrosine kinases, including LYN, FYN, and BLK (Nel et al. 1984, Yamanashi et al. 1991, Flaswinkel and Reth 1994, Saouaf et al. 1994, Hata et al. 1994, Saouaf et al. 1995, reviewed in Gauld and Cambier 2004, reviewed in Harwood and Batista 2010). The protein kinase SYK binds the phosphorylated immunoreceptor tyrosine-activated motifs (ITAMs) on the cytoplasmic tails of Ig-alpha (CD79A, MB-1) and Ig-beta (CD79B, B29)

(Wienands et al. 1995, Rowley et al. 1995, Tsang et al. 2008). The binding causes the activation and autophosphorylation of SYK (Law et al. 1994, Baldock et al. 2000, Irish et al. 2006, Tsang et al. 2008, reviewed in Bradshaw 2010). Activated SYK and other kinases phosphorylate BLNK (SLP-65), BCAP, and CD19 which serve as scaffolds for the assembly of large complexes, the signalosomes, by recruiting phosphoinositol 3-kinase (PI3K), phospholipase C gamma (predominantly PLC-gamma2 in B cells, Coggeshall et al. 1992), NCK, BAM32, BTK, VAV1, and SHC. The effectors are phosphorylated by SYK and other kinases. PLC-gamma associated with BLNK hydrolyzes phosphatidylinositol-4,5-bisphosphate to yield inositol-1,4,5-trisphosphate (IP3) and diacylglycerol (Carter et al. 1991, Kim et al. 2004). IP3 binds receptors on the endoplasmic reticulum and causes release of calcium ions from the ER into the cytosol. The depletion of calcium from the ER in turn activates STIM1 to interact with ORAI and TRPC1 channels in the plasma membrane, resulting in an influx of extracellular calcium ions (Muik et al. 2008, Luik et al. 2008, Park et al. 2009, Mori et al. 2002). PI3K associated with BCAP and CD19 phosphorylates phosphatidylinositol 4,5-bisphosphate to yield phosphatidylinositol 3,4,5-trisphosphate. Second messengers (calcium, diacylglycerol, inositol 1,4,5-trisphosphate, and phosphatidylinositol 3,4,5-trisphosphate) trigger signaling pathways: NF-kappaB is activated via protein kinase C beta, RAS is activated via RasGRP proteins, NF-AT is activated via calcineurin, and AKT (PKB) is activated via PDK1 (reviewed in Shinohara and Kurosaki 2009, Stone 2006).

Module 9

Pathway 1: hsa00061 (KEGG) Fatty acid biosynthesis

NA

Pathway 2: GO:0006633 (GO) fatty acid biosynthetic process

The chemical reactions and pathways resulting in the formation of a fatty acid, any of the aliphatic monocarboxylic acids that can be liberated by hydrolysis from naturally occurring fats and oils. Fatty acids are predominantly straight-chain acids of 4 to 24 carbon atoms, which may be saturated or unsaturated; branched fatty acids and hydroxy fatty acids also occur, and very long chain acids of over 30 carbons are found in waxes.

Pathway 3: GO:0042761 (GO) very long-chain fatty acid biosynthetic process

The chemical reactions and pathways resulting in the formation of a very long-chain fatty acid. A very long-chain fatty acid has an aliphatic tail containing more than 22 carbons.

Pathway 4: GO:0051790 (GO) short-chain fatty acid biosynthetic process

The chemical reactions and pathways resulting in the formation of a short-chain fatty acid. A short-chain fatty acid has an aliphatic tail containing fewer than 6 carbons

Pathway 5: hsa00020 (KEGG) Citrate cycle (TCA cycle)

The citrate cycle (TCA cycle, Krebs cycle) is an important aerobic pathway for the final steps of the oxidation of carbohydrates and fatty acids. The cycle starts with acetyl-CoA, the activated form of acetate, derived from glycolysis and pyruvate oxidation for carbohydrates and from beta oxidation of fatty acids. The two-carbon acetyl group in acetyl-CoA is transferred to the four-carbon compound of oxaloacetate to form the six-carbon compound of citrate. In a series of reactions two carbons in citrate are oxidized to CO₂ and the reaction pathway supplies NADH for use in the oxidative phosphorylation and other metabolic

processes. The pathway also supplies important precursor metabolites including 2-oxoglutarate. At the end of the cycle the remaining four-carbon part is transformed back to oxaloacetate. According to the genome sequence data, many organisms seem to lack genes for the full cycle [MD:M00009], but contain genes for specific segments [MD:M00010 M00011].

Pathway 6: hsa00062 (KEGG) Fatty acid elongation

NA

Pathway 7: R-HSA-75105 (Reactome) Fatty acyl-CoA biosynthesis

Fatty acyl-CoA biosynthesis involves following steps: -Palmitate synthesis catalyzed by Acetyl-CoA carboxylase and Fatty acid synthase -Conversion of palmitic acid to long chain fatty acids and -Conversion of long chain fatty acids to fatty acyl-CoA by acyl-CoA synthases.

Module 10

Pathway 1: hsa04210 (KEGG) Apoptosis

Apoptosis is a genetically programmed process for the elimination of damaged or redundant cells by activation of caspases (aspartate-specific cysteine proteases). The onset of apoptosis is controlled by numerous interrelating processes. The ‘extrinsic’ pathway involves stimulation of members of the tumor necrosis factor (TNF) receptor subfamily, such as TNFR1, CD95/Fas or TRAILR (death receptors), located at the cell surface, by their specific ligands, such as TNF-alpha, FasL or TRAIL, respectively. The ‘intrinsic’ pathway is activated mainly by non-receptor stimuli, such as DNA damage, ER stress, metabolic stress, UV radiation or growth-factor deprivation. The central event in the ‘intrinsic’ pathway is the mitochondrial outer membrane permeabilization (MOMP), which leads to the release of cytochrome c. These two pathways converge at the level of effector caspases, such as caspase-3 and caspase-7. The third major pathway is initiated by the constituents of cytotoxic granules (e.g. Perforin and Granzyme B) that are released by CTLs (cytotoxic T-cells) and NK (natural killer) cells. Granzyme B, similarly to the caspases, cleaves its substrates after aspartic acid residues, suggesting that this protease has the ability to activate members of the caspase family directly. It is the balance between the pro-apoptotic and anti-apoptotic signals that eventually determines whether cells will undergo apoptosis, survive or proliferate. TNF family of ligands activates anti-apoptotic or cell-survival signals as well as apoptotic signals. NGF and Interleukin-3 promotes the survival, proliferation and differentiation of neurons or hematopoietic cells, respectively. Withdrawal of these growth factors leads to cell death, as described above.

Pathway 2: GO:0000165 (GO) MAPK cascade

An intracellular protein kinase cascade containing at least a MAP kinase (MAPK). It starts with the activation of a MAP3K, and the consecutive activation of a MPK2K and a MAPK. The cascade can also contain an additional tier: the upstream MAP4K. The kinases in each tier phosphorylate and activate the kinase in the downstream tier to transmit a signal within a cell.

Pathway 3: GO:0004672 (GO) protein kinase activity

Catalysis of the phosphorylation of an amino acid residue in a protein, usually according to the reaction: a protein + ATP = a phosphoprotein + ADP.

Pathway 4: GO:0004674 (GO) protein serine/threonine kinase activity

Catalysis of the reactions: $\text{ATP} + \text{protein serine} = \text{ADP} + \text{protein serine phosphate}$, and $\text{ATP} + \text{protein threonine} = \text{ADP} + \text{protein threonine phosphate}$.

Pathway 5: GO:0004707 (GO) MAP kinase activity

Catalysis of the reaction: $\text{protein} + \text{ATP} = \text{protein phosphate} + \text{ADP}$. This reaction is the phosphorylation of proteins. Mitogen-activated protein kinase; a family of protein kinases that perform a crucial step in relaying signals from the plasma membrane to the nucleus. They are activated by a wide range of proliferation- or differentiation-inducing signals; activation is strong with agonists such as polypeptide growth factors and tumor-promoting phorbol esters, but weak (in most cell backgrounds) by stress stimuli.

Pathway 6: GO:0070371 (GO) ERK1 and ERK2 cascade

A MAPK cascade containing at least the ERK1 or ERK2 MAP kinases. It starts with the activation of a MAP3K, and the consecutive activation of a MPK2K and of ERK1 or ERK2. The cascade can also contain an additional tier: the upstream MAP4K. The kinases in each tier phosphorylate and activate the kinase in the downstream tier. The ERK1/ERK2 cascade is activated by mitogens, growth factors, G protein-coupled receptors, and results in cellular responses such as cell proliferation, cell differentiation and development.

Pathway 7: hsa04010 (KEGG) MAPK signaling pathway

The mitogen-activated protein kinase (MAPK) cascade is a highly conserved module that is involved in various cellular functions, including cell proliferation, differentiation and migration. Mammals express at least four distinctly regulated groups of MAPKs, extracellular signal-related kinases (ERK)-1/2, Jun amino-terminal kinases (JNK1/2/3), p38 proteins (p38alpha/beta/gamma/delta) and ERK5, that are activated by specific MAPKKs: MEK1/2 for ERK1/2, MKK3/6 for the p38, MKK4/7 (JNKK1/2) for the JNKs, and MEK5 for ERK5. Each MAPKK, however, can be activated by more than one MAPKKK, increasing the complexity and diversity of MAPK signalling. Presumably each MAPKKK confers responsiveness to distinct stimuli. For example, activation of ERK1/2 by growth factors depends on the MAPKKK c-Raf, but other MAPKKKs may activate ERK1/2 in response to pro-inflammatory stimuli.

Module 11

Pathway 1: GO:0051147 (GO) regulation of muscle cell differentiation

Any process that modulates the frequency, rate or extent of muscle cell differentiation.

Pathway 2: GO:0051146 (GO) striated muscle cell differentiation

The process in which a relatively unspecialized cell acquires specialized features of a striated muscle cell; striated muscle fibers are divided by transverse bands into striations, and cardiac and voluntary muscle are types of striated muscle.

Pathway 3: GO:0051148 (GO) negative regulation of muscle cell differentiation

Any process that stops, prevents, or reduces the frequency, rate or extent of muscle cell differentiation.

Pathway 4: GO:0051153 (GO) regulation of striated muscle cell differentiation

Any process that modulates the frequency, rate or extent of striated muscle cell differentiation.

Pathway 5: GO:0051154 (GO) negative regulation of striated muscle cell differentiation

Any process that stops, prevents, or reduces the frequency, rate or extent of striated muscle cell differentiation.

Module 12

Pathway 1: R-HSA-8852135 (Reactome) Protein ubiquitination

Ubiquitin is a small, 76 amino acid residue protein that is conjugated by E3 ubiquitin ligases to other proteins in order to regulate their function or degradation (enzymatic cascade reviewed in Neutzner and Neutzner 2012, Kleiger and Mayor 2014, structures and mechanisms of conjugating enzymes reviewed in Lorenz et al. 2013). Ubiquitination of target proteins usually occurs between the C-terminal glycine residue of ubiquitin and a lysine residue of the target, although linkages with cysteine, serine, and threonine residues are also observed (reviewed in Wang et al. 2012, McDowell and Philpott 2013). Ubiquitin must first be processed from larger precursors and then activated by formation of a thiol ester bond between ubiquitin and an E1 activating enzyme (UBA1 or UBA6) and transfer to an E2 conjugating enzyme before being transferred by an E3 ligase to a target protein. Precursor proteins containing multiple ubiquitin monomers (polyubiquitins) are produced from the UBB and UBC genes; precursors containing a single ubiquitin monomer and a ribosomal protein are produced from the UBA52 and RPS27A genes. Many proteases (deubiquitinases) may potentially process these precursors yielding monomeric ubiquitin. The proteases OTULIN and USP5 are particularly active in cleaving the polyubiquitin precursors, whereas the proteases UCHL3, USP7, and USP9X cleave the ubiquitin-ribosomal protein precursors yielding ubiquitin monomers (Grou et al. 2015). A resultant ubiquitin monomer is activated by adenylation of the C-terminal glycine followed by conjugation of the C-terminus to a cysteine residue of the E1 enzymes UBA1 or UBA6 via a thiol ester bond. The ubiquitin is then transferred from the E1 enzyme to a cysteine residue of one of several E2 enzymes (reviewed in van Wijk and Timmers 2010, Stewart et al. 2016). Through a less well characterized mechanism, E3 ubiquitin ligases then bring a target protein and the E2-ubiquitin conjugate into proximity so that the ubiquitin is transferred via formation of an amide bond to a particular lysine residue (or, in rarer cases, a thiol ester bond to a cysteine residue or an ester bond to a serine or threonine residue) of the target protein (reviewed in Berndsen and Wolberger 2014). Based on protein homologies, families of E3 ubiquitin ligases have been identified that include RING-type ligases (reviewed in Deshaies et al. 2009, Metzger et al. 2012, Metzger et al. 2014), HECT-type ligases (reviewed in Rotin et al. 2009, Metzger et al. 2012), and RBR-type ligases (reviewed in Dove et al. 2016). A subset of the RING-type ligases participate in CULLIN-RING ligase complexes (CRLs which include SCF complexes, reviewed in Lee and Zhou 2007, Genschik et al. 2013, Skaar et al. 2013, Lee et al. 2014). Some E3-E2 combinations catalyze mono-ubiquitination of the target protein (reviewed in Nakagawa and Nakayama 2015). Other E3-E2 combinations catalyze conjugation of further ubiquitin monomers to the initial ubiquitin, forming polyubiquitin chains. (It may also be possible for some E3-E2 combinations to preassemble polyubiquitin and transfer it as a unit to the target protein.) Ubiquitin contains several lysine (K) residues and a free alpha amino group to which further ubiquitin can be conjugated. Thus different types of polyubiquitin are possible: K11 linked polyubiquitin is observed in endoplasmic reticulum-associated degradation (ERAD), K29 linked polyubiquitin is observed in lysosomal degradation, K48 linked polyubiquitin directs target proteins to the proteasome for degradation, whereas K63 linked polyubiquitin generally acts as a scaffold to recruit other proteins in several cellular processes, notably DNA repair (reviewed in Komander et al. 2009). Ubiquitination is highly regulated (reviewed in Vittal et al. 2015) and affects all cellular processes including DNA damage response (reviewed in Brown and Jackson 2015), immune signaling (reviewed in

Park et al. 2014, Lutz-Nicoladoni et al. 2015), and regulation of normal and cancerous cell growth (reviewed in Skaar and Pagano 2009, Yerlikaya and Yontem 2013, Strikoudis et al. 2014).

Pathway 2: hsa04120 (KEGG) Ubiquitin mediated proteolysis

Protein ubiquitination plays an important role in eukaryotic cellular processes. It mainly functions as a signal for 26S proteasome dependent protein degradation. The addition of ubiquitin to proteins being degraded is performed by a reaction cascade consisting of three enzymes, named E1 (ubiquitin activating enzyme), E2 (ubiquitin conjugating enzyme), and E3 (ubiquitin ligase). Each E3 has specificity to its substrate, or proteins to be targeted by ubiquitination. Many E3s are discovered in eukaryotes and they are classified into four types: HECT type, U-box type, single RING-finger type, and multi-subunit RING-finger type. Multi-subunit RING-finger E3s are exemplified by cullin-Rbx E3s and APC/C. They consist of a RING-finger-containing subunit (RBX1 or RBX2) that functions to bind E2s, a scaffold-like cullin molecule, adaptor proteins, and a target recognizing subunit that binds substrates.

Pathway 3: hsa03050 (KEGG) Proteasome

The proteasome is a protein-destroying apparatus involved in many essential cellular functions, such as regulation of cell cycle, cell differentiation, signal transduction pathways, antigen processing for appropriate immune responses, stress signaling, inflammatory responses, and apoptosis. It is capable of degrading a variety of cellular proteins in a rapid and timely fashion and most substrate proteins are modified by ubiquitin before their degradation by the proteasome. The proteasome is a large protein complex consisting of a proteolytic core called the 20S particle and ancillary factors that regulate its activity in various ways. The most common form is the 26S proteasome containing one 20S core particle and two 19S regulatory particles that enable the proteasome to degrade ubiquitinated proteins by an ATP-dependent mechanism. Another form is the immunoproteasome containing two 11S regulatory particles, PA28 alpha and PA28 beta, which are induced by interferon gamma under the conditions of intensified immune response. Other regulatory particles include PA28 gamma and PA200. Although PA28 gamma also belongs to a family of activators of the 20S proteasome, it is localized within the nucleus and forms a homoheptamer. PA28 gamma has been implicated in the regulation of cell cycle progression and apoptosis. PA200 has been identified as a large nuclear protein that stimulates proteasomal hydrolysis of peptides.

Pathway 4: GO:0016567 (GO) protein ubiquitination

The process in which one or more ubiquitin groups are added to a protein.

Pathway 5: GO:0061630 (GO) ubiquitin protein ligase activity

Catalysis of the transfer of ubiquitin to a substrate protein via the reaction $X\text{-ubiquitin} + S = X + S\text{-ubiquitin}$, where X is either an E2 or E3 enzyme, the X-ubiquitin linkage is a thioester bond, and the S-ubiquitin linkage is an amide bond: an isopeptide bond between the C-terminal glycine of ubiquitin and the epsilon-amino group of lysine residues in the substrate or, in the linear extension of ubiquitin chains, a peptide bond between the C-terminal glycine and N-terminal methionine of ubiquitin residues.

Supplementary Note 8. Random functional module set

Random Module 4

Pathway 1: R-HSA-68952 (Reactome) DNA replication initiation

DNA polymerases are not capable of de novo DNA synthesis and require synthesis of a primer, usually by a DNA-dependent RNA polymerase (primase) to begin DNA synthesis. In eukaryotic cells, the primer is synthesized by DNA polymerase alpha:primase. First, the DNA primase portion of this complex synthesizes approximately 6-10 nucleotides of RNA primer and then the DNA polymerase portion synthesizes an additional 20 nucleotides of DNA (Frick & Richardson 2002; Wang et al 1984).

Pathway 2: GO:0051790 (GO) short-chain fatty acid biosynthetic process

The chemical reactions and pathways resulting in the formation of a short-chain fatty acid. A short-chain fatty acid has an aliphatic tail containing fewer than 6 carbons

Pathway 3: GO:0051147 (GO) regulation of muscle cell differentiation

Any process that modulates the frequency, rate or extent of muscle cell differentiation.

Random Module 5

Pathway 1: GO:0045251 (GO) electron transfer flavoprotein complex

A protein complex facilitating the electron transfer from an acyl-CoA molecule to ubiquinone via its flavin adenine dinucleotide (FAD) cofactor. Usually contains an alpha and a beta subunit and the structural cofactor adenosine monophosphate (AMP). Part of a system that oxidizes an acyl-CoA molecule and reduces ubiquinone and other acceptors in the electron transport system.

Pathway 2: GO:0006955 (GO) immune response

Any immune system process that functions in the calibrated response of an organism to a potential internal or invasive threat.

Pathway 3: GO:0051154 (GO) negative regulation of striated muscle cell differentiation

Any process that stops, prevents, or reduces the frequency, rate or extent of striated muscle cell differentiation.

Random Module 7

Pathway 1: hsa03030 (KEGG) DNA replication

A complex network of interacting proteins and enzymes is required for DNA replication. Generally, DNA replication follows a multistep enzymatic pathway. At the DNA replication fork, a DNA helicase (DnaB or MCM complex) precedes the DNA synthetic machinery and unwinds the duplex parental DNA in cooperation with the SSB or RPA. On the leading strand, replication occurs continuously in a 5 to 3 direction, whereas on the lagging strand, DNA replication occurs discontinuously by synthesis and joining of short Okazaki fragments. In prokaryotes, the leading strand replication apparatus consists of a DNA polymerase (pol III core), a sliding clamp (beta), and a clamp loader (gamma delta complex). The DNA primase (DnaG) is needed to form RNA primers. Normally, during replication of the lagging-strand DNA template, an RNA primer is removed either by an RNase H or by the 5 to 3 exonuclease activity of DNA pol I, and the DNA ligase joins the Okazaki fragments. In eukaryotes, three DNA polymerases (alpha, delta, and epsilon) have been identified. DNA primase forms a permanent complex with DNA polymerase alpha. PCNA and RFC function as a clamp and a clamp loader. FEN 1 and RNase H1 remove the RNA from the Okazaki fragments and DNA ligase I joins the DNA.

Pathway 2: hsa00061 (KEGG) Fatty acid biosynthesis
NA

Pathway 3: GO:0016567 (GO) protein ubiquitination
The process in which one or more ubiquitin groups are added to a protein.

Random Module 8

Pathway 1: hsa04724 (KEGG) Glutamatergic synapse

Glutamate is the major excitatory neurotransmitter in the mammalian central nervous system(CNS). Glutamate is packaged into synaptic vesicles in the presynaptic terminal. Once released into the synaptic cleft, glutamate acts on postsynaptic ionotropic glutamate receptors (iGluRs) to mediate fast excitatory synaptic transmission. Glutamate can also act on metabotropic glutamate receptors (mGluRs) and exert a variety of modulatory effects through their coupling to G proteins and the subsequent recruitment of second messenger systems. Presynaptically localized Group II and Group III mGluRs are thought to represent the classical inhibitory autoreceptor mechanism that suppresses excess glutamate release. After its action on these receptors, glutamate can be removed from the synaptic cleft by EAATs located either on the presynaptic terminal, neighboring glial cells, or the postsynaptic neuron. In glia, glutamate is converted to glutamine, which is then transported back to the presynaptic terminal and converted back to glutamate.

Pathway 2: GO:0009055 (GO) electron transfer activity

A molecular function representing the directed movement of electrons from one molecular entity to another, typically mediated by electron carriers or acceptors, resulting in the transfer of energy and/or the reduction-oxidation (redox) transformation of chemical species. This activity is fundamental to various biological processes, including cellular respiration and photosynthesis, as well as numerous enzymatic reactions involved in metabolic pathways.

Pathway 3: GO:0002250 (GO) adaptive immune response

An immune response mediated by cells expressing specific receptors for antigens produced through a somatic diversification process, and allowing for an enhanced secondary response to subsequent exposures to the same antigen (immunological memory).

Pathway 4: hsa04662 (KEGG) B cell receptor signaling pathway

B cells are an important component of adaptive immunity. They produce and secrete millions of different antibody molecules, each of which recognizes a different (foreign) antigen. The B cell receptor (BCR) is an integral membrane protein complex that is composed of two immunoglobulin (Ig) heavy chains, two Ig light chains and two heterodimers of Ig-alpha and Ig-beta. After BCR ligation by antigen, three main protein tyrosine kinases (PTKs) -the SRC-family kinase LYN, SYK and the TEC-family kinase BTK- are activated. Phosphatidylinositol 3-kinase (PI3K) and phospholipase C-gamma 2 (PLC-gamma 2) are important downstream effectors of BCR signalling. This signalling ultimately results in the expression of immediate early genes that further activate the expression of other genes involved in B cell proliferation, differentiation and Ig production as well as other processes.

Pathway 5: R-HSA-388841 (Reactome) Regulation of T cell activation by CD28 family

Optimal activation of T lymphocytes requires a carefully orchestrated interplay between two key signals. The first signal originates from the T cell receptor (TCR) recognizing a specific antigen. However, this initial encounter is insufficient on its own. A crucial “costimulatory” signal is needed for full T cell activation, delivered by the engagement of costimulatory receptors, such as CD28, on the T cell surface with their corresponding ligands on antigen-presenting cells (APCs) (Chen & Flies 2013, Acuto et al. 2003, Slavik et al. 1999). The CD28 superfamily is central to costimulation, encompassing a diverse group of receptors including CD28, CTLA-4, ICOS, PD-1, and BTLA. These receptors have both positive and negative influences on T cell activation. CD28 and ICOS provide a positive boost, while CTLA-4, PD-1, and BTLA act as inhibitory brakes. CD28 and CTLA-4 bind to B7 family ligands CD80 (B7.1) and CD86 (B7.2), with CD28 delivering stimulatory signals and CTLA-4 providing inhibitory signals. ICOS interacts with ICOS ligand (ICOS-L), enhancing T cell activation and function. PD-1 binds to PD-L1 and PD-L2, mediating inhibitory signals that modulate the immune response. BTLA engages with HVEM (Herpesvirus entry mediator), transmitting inhibitory signals. This balance between stimulatory and inhibitory signals is essential, allowing the immune system to mount effective responses against pathogens while preventing the induction of T cell unresponsiveness and apoptosis (Chen & Flies 2013, Sharpe & Freeman 2002, Carreno & Collins 2002). The expression of these receptors varies: CD28 is constantly present on naive T cells, whereas CTLA-4 expression depends on prior CD28 engagement. ICOS, PD-1, and BTLA are induced only after initial T cell stimulation. These variations in expression ensure precise regulation of T cell responses at different stages of activation (Sharpe & Freeman 2002). The receptors themselves also exhibit structural differences. CD28, CTLA-4, and ICOS share a single extracellular domain resembling an immunoglobulin molecule (IgV-like). In contrast, BTLA has a distinct structure with an IgC-like domain. Similarly, the costimulatory ligands, such as B7-1 and B7-2, have unique structural features that enable specific interactions with their partner receptors.

Pathway 6: GO:0042761 (GO) very long-chain fatty acid biosynthetic process

The chemical reactions and pathways resulting in the formation of a very long-chain fatty acid. A very long-chain fatty acid has an aliphatic tail containing more than 22 carbons.

Pathway 7: R-HSA-75105 (Reactome) Fatty acyl-CoA biosynthesis

Fatty acyl-CoA biosynthesis involves following steps: -Palmitate synthesis catalyzed by Acetyl-CoA carboxylase and Fatty acid synthase -Conversion of palmitic acid to long chain fatty acids and -Conversion of long chain fatty acids to fatty acyl-CoA by acyl-CoA synthases.

Random Module 9

Pathway 1: hsa04210 (KEGG) Apoptosis

Apoptosis is a genetically programmed process for the elimination of damaged or redundant cells by activation of caspases (aspartate-specific cysteine proteases). The onset of apoptosis is controlled by numerous interrelating processes. The ‘extrinsic’ pathway involves stimulation of members of the tumor necrosis factor (TNF) receptor subfamily, such as TNFR1, CD95/Fas or TRAILR (death receptors), located at the cell surface, by their specific ligands, such as TNF- α , FasL or TRAIL, respectively. The ‘intrinsic’ pathway is activated mainly by non-receptor stimuli, such as DNA damage, ER stress, metabolic stress, UV radiation or growth-factor deprivation. The central event in the ‘intrinsic’ pathway is the mitochondrial outer membrane permeabilization (MOMP), which leads to the release of cytochrome c. These two pathways converge at the level of effector caspases, such as caspase-3 and caspase-7. The third major

pathway is initiated by the constituents of cytotoxic granules (e.g. Perforin and Granzyme B) that are released by CTLs (cytotoxic T-cells) and NK (natural killer) cells. Granzyme B, similarly to the caspases, cleaves its substrates after aspartic acid residues, suggesting that this protease has the ability to activate members of the caspase family directly. It is the balance between the pro-apoptotic and anti-apoptotic signals that eventually determines whether cells will undergo apoptosis, survive or proliferate. TNF family of ligands activates anti-apoptotic or cell-survival signals as well as apoptotic signals. NGF and Interleukin-3 promotes the survival, proliferation and differentiation of neurons or hematopoietic cells, respectively. Withdrawal of these growth factors leads to cell death, as described above.

Pathway 2: GO:0004672 (GO) protein kinase activity

Catalysis of the phosphorylation of an amino acid residue in a protein, usually according to the reaction: a protein + ATP = a phosphoprotein + ADP.

Pathway 3: GO:0098985 (GO) asymmetric, glutamatergic, excitatory synapse

A neuron to neuron synapse with a postsynaptic density, that uses glutamate as a neurotransmitter and whose activity results in excitatory postsynaptic potentials.

Pathway 4: R-HSA-1280218 (Reactome) Adaptive Immune System

Adaptive immunity refers to antigen-specific immune response efficiently involved in clearing the pathogens. The adaptive immune system is comprised of B and T lymphocytes that express receptors with remarkable diversity tailored to recognize aspects of particular pathogens or antigens. During infection, dendritic cells (DC) which act as sentinels in the peripheral tissues recognize and pick up the pathogen in the form of antigenic determinants and then process these antigens and present them to T cells. These T cells of appropriate specificity respond to the antigen, and either kill the pathogen directly or secrete cytokines that will stimulate B lymphocyte response. B cells provide humoral immunity by secreting antibodies specific for the pathogen or antigen.

Pathway 5: GO:0051146 (GO) striated muscle cell differentiation

The process in which a relatively unspecialized cell acquires specialized features of a striated muscle cell; striated muscle fibers are divided by transverse bands into striations, and cardiac and voluntary muscle are types of striated muscle.

Pathway 6: hsa03050 (KEGG) Proteasome

The proteasome is a protein-destroying apparatus involved in many essential cellular functions, such as regulation of cell cycle, cell differentiation, signal transduction pathways, antigen processing for appropriate immune responses, stress signaling, inflammatory responses, and apoptosis. It is capable of degrading a variety of cellular proteins in a rapid and timely fashion and most substrate proteins are modified by ubiquitin before their degradation by the proteasome. The proteasome is a large protein complex consisting of a proteolytic core called the 20S particle and ancillary factors that regulate its activity in various ways. The most common form is the 26S proteasome containing one 20S core particle and two 19S regulatory particles that enable the proteasome to degrade ubiquitinated proteins by an ATP-dependent mechanism. Another form is the immunoproteasome containing two 11S regulatory particles, PA28 alpha and PA28 beta, which are induced by interferon gamma under the conditions of intensified immune response. Other regulatory particles include PA28 gamma and PA200. Although PA28 gamma also belongs

to a family of activators of the 20S proteasome, it is localized within the nucleus and forms a homoheptamer. PA28 gamma has been implicated in the regulation of cell cycle progression and apoptosis. PA200 has been identified as a large nuclear protein that stimulates proteasomal hydrolysis of peptides.

Pathway 7: GO:0061630 (GO) ubiquitin protein ligase activity

Catalysis of the transfer of ubiquitin to a substrate protein via the reaction $X\text{-ubiquitin} + S = X + S\text{-ubiquitin}$, where X is either an E2 or E3 enzyme, the X-ubiquitin linkage is a thioester bond, and the S-ubiquitin linkage is an amide bond: an isopeptide bond between the C-terminal glycine of ubiquitin and the epsilon-amino group of lysine residues in the substrate or, in the linear extension of ubiquitin chains, a peptide bond between the C-terminal glycine and N-terminal methionine of ubiquitin residues.

Random Module 10

Pathway 1: GO:0004674 (GO) protein serine/threonine kinase activity

Catalysis of the reactions: $ATP + \text{protein serine} = ADP + \text{protein serine phosphate}$, and $ATP + \text{protein threonine} = ADP + \text{protein threonine phosphate}$.

Pathway 2: GO:0070371 (GO) ERK1 and ERK2 cascade

A MAPK cascade containing at least the ERK1 or ERK2 MAP kinases. It starts with the activation of a MAP3K, and the consecutive activation of a MPK2K and of ERK1 or ERK2. The cascade can also contain an additional tier: the upstream MAP4K. The kinases in each tier phosphorylate and activate the kinase in the downstream tier. The ERK1/ERK2 cascade is activated by mitogens, growth factors, G protein-coupled receptors, and results in cellular responses such as cell proliferation, cell differentiation and development.

Pathway 3: hsa04010 (KEGG) MAPK signaling pathway

The mitogen-activated protein kinase (MAPK) cascade is a highly conserved module that is involved in various cellular functions, including cell proliferation, differentiation and migration. Mammals express at least four distinctly regulated groups of MAPKs, extracellular signal-related kinases (ERK)-1/2, Jun amino-terminal kinases (JNK1/2/3), p38 proteins (p38alpha/beta/gamma/delta) and ERK5, that are activated by specific MAPKKs: MEK1/2 for ERK1/2, MKK3/6 for the p38, MKK4/7 (JNKK1/2) for the JNKs, and MEK5 for ERK5. Each MAPKK, however, can be activated by more than one MAPKKK, increasing the complexity and diversity of MAPK signalling. Presumably each MAPKKK confers responsiveness to distinct stimuli. For example, activation of ERK1/2 by growth factors depends on the MAPKKK c-Raf, but other MAPKKKs may activate ERK1/2 in response to pro-inflammatory stimuli.

Pathway 4: GO:0098978 (GO) glutamatergic synapse

A synapse that uses glutamate as a neurotransmitter.

Pathway 5: GO:0006633 (GO) fatty acid biosynthetic process

The chemical reactions and pathways resulting in the formation of a fatty acid, any of the aliphatic monocarboxylic acids that can be liberated by hydrolysis from naturally occurring fats and oils. Fatty acids are predominantly straight-chain acids of 4 to 24 carbon atoms, which may be saturated or unsaturated; branched fatty acids and hydroxy fatty acids also occur, and very long chain acids of over 30 carbons are found in waxes.

Pathway 6: GO:0051153 (GO) regulation of striated muscle cell differentiation

Any process that modulates the frequency, rate or extent of striated muscle cell differentiation.

Pathway 7: hsa04120 (KEGG) Ubiquitin mediated proteolysis

Protein ubiquitination plays an important role in eukaryotic cellular processes. It mainly functions as a signal for 26S proteasome dependent protein degradation. The addition of ubiquitin to proteins being degraded is performed by a reaction cascade consisting of three enzymes, named E1 (ubiquitin activating enzyme), E2 (ubiquitin conjugating enzyme), and E3 (ubiquitin ligase). Each E3 has specificity to its substrate, or proteins to be targeted by ubiquitination. Many E3s are discovered in eukaryotes and they are classified into four types: HECT type, U-box type, single RING-finger type, and multi-subunit RING-finger type. Multi-subunit RING-finger E3s are exemplified by cullin-Rbx E3s and APC/C. They consist of a RING-finger-containing subunit (RBX1 or RBX2) that functions to bind E2s, a scaffold-like cullin molecule, adaptor proteins, and a target recognizing subunit that binds substrates.

Random Module 11

Pathway 1: GO:0004707 (GO) MAP kinase activity

Catalysis of the reaction: protein + ATP = protein phosphate + ADP. This reaction is the phosphorylation of proteins. Mitogen-activated protein kinase; a family of protein kinases that perform a crucial step in relaying signals from the plasma membrane to the nucleus. They are activated by a wide range of proliferation- or differentiation-inducing signals; activation is strong with agonists such as polypeptide growth factors and tumor-promoting phorbol esters, but weak (in most cell backgrounds) by stress stimuli.

Pathway 2: R-HSA-69306 (Reactome) DNA Replication

Studies in the past decade have suggested that the basic mechanism of DNA replication initiation is conserved in all kingdoms of life. Initiation in unicellular eukaryotes, in particular *Saccharomyces cerevisiae* (budding yeast), is well understood, and has served as a model for studies of DNA replication initiation in multicellular eukaryotes, including humans. In general terms, the first step of initiation is the binding of the replication initiator to the origin of replication. The replicative helicase is then assembled onto the origin, usually by a helicase assembly factor. Either shortly before or shortly after helicase assembly, some local unwinding of the origin of replication occurs in a region rich in adenine and thymine bases (often termed a DNA unwinding element, DUE). The unwound region provides the substrate for primer synthesis and initiation of DNA replication. The best-defined eukaryotic origins are those of *S. cerevisiae*, which have well-conserved sequence elements for initiator binding, DNA unwinding and binding of accessory proteins. In multicellular eukaryotes, unlike *S. cerevisiae*, these loci appear not to be defined by the presence of a DNA sequence motif. Indeed, choice of replication origins in a multicellular eukaryote may vary with developmental stage and tissue type. In cell-free models of metazoan DNA replication, such as the one provided by *Xenopus* egg extracts, there are only limited DNA sequence specificity requirements for replication initiation (Kelly & Brown 2000; Bell & Dutta 2002; Marahrens & Stillman 1992; Cimbara & Groudine 2001; Mahbubani et al 1992, Hyrien & Mechali 1993).

Pathway 3: R-HSA-983705 (Reactome) Signaling by the B Cell Receptor (BCR)

Mature B cells express IgM and IgD immunoglobulins which are complexed at the plasma membrane with Ig-alpha (CD79A, MB-1) and Ig-beta (CD79B, B29) to form the B cell receptor (BCR) (Fu et al. 1974, Fu et al. 1975, Kunkel et al. 1975, Van Noesel et al. 1992, Sanchez et al. 1993, reviewed in Brezski and Monroe

2008). Binding of antigen to the immunoglobulin activates phosphorylation of immunoreceptor tyrosine-based activation motifs (ITAMs) in the cytoplasmic tails of Ig-alpha and Ig-beta by Src family tyrosine kinases, including LYN, FYN, and BLK (Nel et al. 1984, Yamanashi et al. 1991, Flaswinkel and Reth 1994, Saouaf et al. 1994, Hata et al. 1994, Saouaf et al. 1995, reviewed in Gauld and Cambier 2004, reviewed in Harwood and Batista 2010). The protein kinase SYK binds the phosphorylated immunoreceptor tyrosine-activated motifs (ITAMs) on the cytoplasmic tails of Ig-alpha (CD79A, MB-1) and Ig-beta (CD79B, B29) (Wienands et al. 1995, Rowley et al. 1995, Tsang et al. 2008). The binding causes the activation and autophosphorylation of SYK (Law et al. 1994, Baldock et al. 2000, Irish et al. 2006, Tsang et al. 2008, reviewed in Bradshaw 2010). Activated SYK and other kinases phosphorylate BLNK (SLP-65), BCAP, and CD19 which serve as scaffolds for the assembly of large complexes, the signalosomes, by recruiting phosphoinositol 3-kinase (PI3K), phospholipase C gamma (predominantly PLC-gamma2 in B cells, Coggeshall et al. 1992), NCK, BAM32, BTK, VAV1, and SHC. The effectors are phosphorylated by SYK and other kinases. PLC-gamma associated with BLNK hydrolyzes phosphatidylinositol-4,5-bisphosphate to yield inositol-1,4,5-trisphosphate (IP3) and diacylglycerol (Carter et al. 1991, Kim et al. 2004). IP3 binds receptors on the endoplasmic reticulum and causes release of calcium ions from the ER into the cytosol. The depletion of calcium from the ER in turn activates STIM1 to interact with ORAI and TRPC1 channels in the plasma membrane, resulting in an influx of extracellular calcium ions (Muik et al. 2008, Luik et al. 2008, Park et al. 2009, Mori et al. 2002). PI3K associated with BCAP and CD19 phosphorylates phosphatidylinositol 4,5-bisphosphate to yield phosphatidylinositol 3,4,5-trisphosphate. Second messengers (calcium, diacylglycerol, inositol 1,4,5-trisphosphate, and phosphatidylinositol 3,4,5-trisphosphate) trigger signaling pathways: NF-kappaB is activated via protein kinase C beta, RAS is activated via RasGRP proteins, NF-AT is activated via calcineurin, and AKT (PKB) is activated via PDK1 (reviewed in Shinohara and Kurosaki 2009, Stone 2006).

Pathway 4: hsa00062 (KEGG) Fatty acid elongation
NA

Pathway 5: GO:0051148 (GO) negative regulation of muscle cell differentiation
Any process that stops, prevents, or reduces the frequency, rate or extent of muscle cell differentiation.

Random Module 12

Pathway 1: GO:0000165 (GO) MAPK cascade

An intracellular protein kinase cascade containing at least a MAP kinase (MAPK). It starts with the activation of a MAP3K, and the consecutive activation of a MPK2K and a MAPK. The cascade can also contain an additional tier: the upstream MAP4K. The kinases in each tier phosphorylate and activate the kinase in the downstream tier to transmit a signal within a cell.

Pathway 2: hsa00190 (KEGG) Oxidative phosphorylation
NA

Pathway 3: hsa04660 (KEGG) T cell receptor signaling pathway

Activation of T lymphocytes is a key event for an efficient response of the immune system. It requires the involvement of the T-cell receptor (TCR) as well as costimulatory molecules such as CD28. Engagement of these receptors through the interaction with a foreign antigen associated with major histocompatibility

complex molecules and CD28 counter-receptors B7.1/B7.2, respectively, results in a series of signaling cascades. These cascades comprise an array of protein-tyrosine kinases, phosphatases, GTP-binding proteins and adaptor proteins that regulate generic and specialised functions, leading to T-cell proliferation, cytokine production and differentiation into effector cells.

Pathway 4: hsa00020 (KEGG) Citrate cycle (TCA cycle)

The citrate cycle (TCA cycle, Krebs cycle) is an important aerobic pathway for the final steps of the oxidation of carbohydrates and fatty acids. The cycle starts with acetyl-CoA, the activated form of acetate, derived from glycolysis and pyruvate oxidation for carbohydrates and from beta oxidation of fatty acids. The two-carbon acetyl group in acetyl-CoA is transferred to the four-carbon compound of oxaloacetate to form the six-carbon compound of citrate. In a series of reactions two carbons in citrate are oxidized to CO₂ and the reaction pathway supplies NADH for use in the oxidative phosphorylation and other metabolic processes. The pathway also supplies important precursor metabolites including 2-oxoglutarate. At the end of the cycle the remaining four-carbon part is transformed back to oxaloacetate. According to the genome sequence data, many organisms seem to lack genes for the full cycle [MD:M00009], but contain genes for specific segments [MD:M00010 M00011].

Pathway 5: R-HSA-8852135 (Reactome) Protein ubiquitination

Ubiquitin is a small, 76 amino acid residue protein that is conjugated by E3 ubiquitin ligases to other proteins in order to regulate their function or degradation (enzymatic cascade reviewed in Neutzner and Neutzner 2012, Kleiger and Mayor 2014, structures and mechanisms of conjugating enzymes reviewed in Lorenz et al. 2013). Ubiquitination of target proteins usually occurs between the C-terminal glycine residue of ubiquitin and a lysine residue of the target, although linkages with cysteine, serine, and threonine residues are also observed (reviewed in Wang et al. 2012, McDowell and Philpott 2013). Ubiquitin must first be processed from larger precursors and then activated by formation of a thiol ester bond between ubiquitin and an E1 activating enzyme (UBA1 or UBA6) and transfer to an E2 conjugating enzyme before being transferred by an E3 ligase to a target protein. Precursor proteins containing multiple ubiquitin monomers (polyubiquitins) are produced from the UBB and UBC genes; precursors containing a single ubiquitin monomer and a ribosomal protein are produced from the UBA52 and RPS27A genes. Many proteases (deubiquitinases) may potentially process these precursors yielding monomeric ubiquitin. The proteases OTULIN and USP5 are particularly active in cleaving the polyubiquitin precursors, whereas the proteases UCHL3, USP7, and USP9X cleave the ubiquitin-ribosomal protein precursors yielding ubiquitin monomers (Grou et al. 2015). A resultant ubiquitin monomer is activated by adenylation of the C-terminal glycine followed by conjugation of the C-terminus to a cysteine residue of the E1 enzymes UBA1 or UBA6 via a thiol ester bond. The ubiquitin is then transferred from the E1 enzyme to a cysteine residue of one of several E2 enzymes (reviewed in van Wijk and Timmers 2010, Stewart et al. 2016). Through a less well characterized mechanism, E3 ubiquitin ligases then bring a target protein and the E2-ubiquitin conjugate into proximity so that the ubiquitin is transferred via formation of an amide bond to a particular lysine residue (or, in rarer cases, a thiol ester bond to a cysteine residue or an ester bond to a serine or threonine residue) of the target protein (reviewed in Berndsen and Wolberger 2014). Based on protein homologies, families of E3 ubiquitin ligases have been identified that include RING-type ligases (reviewed in Deshaies et al. 2009, Metzger et al. 2012, Metzger et al. 2014), HECT-type ligases (reviewed in Rotin et al. 2009, Metzger et al. 2012), and RBR-type ligases (reviewed in Dove et al. 2016). A subset of the RING-type ligases participate in CULLIN-RING ligase complexes (CRLs which include SCF complexes, reviewed in

Lee and Zhou 2007, Genschik et al. 2013, Skaar et al. 2013, Lee et al. 2014). Some E3-E2 combinations catalyze mono-ubiquitination of the target protein (reviewed in Nakagawa and Nakayama 2015). Other E3-E2 combinations catalyze conjugation of further ubiquitin monomers to the initial ubiquitin, forming polyubiquitin chains. (It may also be possible for some E3-E2 combinations to preassemble polyubiquitin and transfer it as a unit to the target protein.) Ubiquitin contains several lysine (K) residues and a free alpha amino group to which further ubiquitin can be conjugated. Thus different types of polyubiquitin are possible: K11 linked polyubiquitin is observed in endoplasmic reticulum-associated degradation (ERAD), K29 linked polyubiquitin is observed in lysosomal degradation, K48 linked polyubiquitin directs target proteins to the proteasome for degradation, whereas K63 linked polyubiquitin generally acts as a scaffold to recruit other proteins in several cellular processes, notably DNA repair (reviewed in Komander et al. 2009). Ubiquitination is highly regulated (reviewed in Vittal et al. 2015) and affects all cellular processes including DNA damage response (reviewed in Brown and Jackson 2015), immune signaling (reviewed in Park et al. 2014, Lutz-Nicoladoni et al. 2015), and regulation of normal and cancerous cell growth (reviewed in Skaar and Pagano 2009, Yerlikaya and Yontem 2013, Strikoudis et al. 2014).

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